

Benders decomposition applied to profit maximizing hub location problem with incomplete hub network

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ABSTRACT

This paper addresses a profit maximizing multiple allocation hub network design problem, which is a hub location problem with incomplete hub network, aiming to determine the quantity and location of the hubs, allocate demand nodes to these hubs, establish the network topology, and route the demand flows to satisfy the demands between the origin and destination pairs that were selected to be served in order to obtain the highest profit. This problem also considers the multiple allocation strategy, does not allow direct connections between non-hub nodes, does not impose capacity constraints, and assumes that there are fixed costs involved in installing hubs and installing hub arcs. A new formulation for the problem is presented and it is proved that this formulation is stronger than another one in the literature, which models the same problem. Several versions of the Benders decomposition method, combining the generation of Pareto-optimal cuts, multiple cuts strategies, and Benders branch-and-cut scheme, are proposed and analyzed through extensive computational experiments on benchmark instances. The best-proposed Benders decomposition versions were compared with the CPLEX solver, with and without the activation of its Benders decomposition algorithm. The results showed that these algorithms perform much better than the solver versions. The proposed Benders decomposition approach can solve benchmark instances with up to 150 nodes.

1. Introduction

Hubs are kinds of facilities responsible for collecting, consolidating, transferring, and distributing flows in a network, connecting the origin and destination pairs. The demand between the pairs of nodes is routed through the hubs instead of using direct connections. Due to the consolidation of flows in hub nodes, economies of scale in the transportation cost can be achieved.

The classical hub location problems (HLPs) can be defined as location and allocation problems, consisting of locating hubs, allocating demand nodes to these hubs, and routing the flows between the origin and destination pairs to optimize a given objective. Except in specific cases, the HLPs are NP-hard problems. The difficulty in solving them arises from the combination of location and allocation decisions (Campbell and O'Kelly, 2012).

In the last decades, the HLPs have been applied in many areas, such as passengers and cargo transportation systems (Jaillet et al., 1996; Tan and Kara, 2007; Wieberneit, 2008; Lin et al., 2012), telecommunication networks (Carello et al., 2004; Bollapragada et al., 2006; Kim and O'Kelly, 2009), and postal delivery services (Kuby and Gray, 1993; Çetiner et al., 2010).

The hubs and the hub arcs (arcs connecting two hubs) form the hub-level network, while the non-hub nodes and the arcs connecting the non-hub nodes to the hubs (access arcs) compose the access network. In general, the topology of these networks is closely related to the application area and constitutes one of the main classification criteria for the HLPs.

O'Kelly and Miller (1994) proposed eight classification protocols for this class of problems. These protocols are based on three topological characteristics of a network: (i) allocation type, where non-hub nodes are assigned to a single hub (single allocation) or can be connected to more than one hub (multiple allocation); (ii) allowing or not allowing direct connections between non-hub nodes; and (iii) inter-connectivity between the hub nodes, in which every pair of hubs is interconnected (complete hub network) or is partially interconnected (incomplete hub network). Additionally to the single and multiple allocation strategies, the r -allocation strategy, introduced by Yaman (2011), allows each node to be assigned to a maximum of r hubs.

Three additional characteristics of HLPs are (i) the prior fixing of the number of hubs in the network (giving rise to the p -hub median problem, for instance); (ii) the limitation of the hub and hub arc

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capacities (yielding the capacitated or uncapacitated problems); and (iii) the existence of fixed costs for establishing hubs and hub arcs. From these characteristics, the literature shows different variants of HLPs. Campbell and O’Kelly (2012), Farahani et al. (2013) and Contreras (2015), among others, present reviews of these many variants of the problem.

Most HLPs are modeled mainly from a cost-oriented objective, having, as the main objective, to minimize the costs to satisfy all the demand. However, as mentioned by Taherkhani and Alumur (2018), from the point of view of profit, it may be more advantageous not to meet the demand of some pairs of origin and destination nodes, especially if the cost is greater than the revenue associated with meeting this demand. In this situation, it is more appropriate to maximize profit instead of minimizing cost. According to Alibeyg et al. (2016), HLPs focused on profit maximization integrate additional strategic decisions about the network configuration and demands that have to be served and address the trade-off between the revenue of the served pairs of nodes and the costs associated with the network design and transportation of demand flows. Broadly speaking, this problem has the following strategic decisions: (i) where to locate the hubs; (ii) what edges to activate; and (iii) which pairs of nodes will be serviced. The operational decisions determine how to route the demands that are selected to be served. This problem generalizes classic HLPs as they incorporate one additional level to the decision-making process. Therefore, profit maximization is more complex than cost minimization; however, this is the real configuration issue in many real-life hub networks (Taherkhani et al., 2020).

Furthermore, most of the classic works deal with a structure in which the hubs are fully interconnected. However, there are situations in which this assumption is not adequate, especially if the costs to activate hub arcs are very high or, on the other hand, the demand flow between some hub arcs is low. In this case, the activation of all arcs between hubs can be impractical, not compensating for a possible reduction in transportation costs that could be achieved with a complete hub network; thus, an incomplete hub network may be more suitable. Incomplete hub networks are often found in public transport systems, cargo transport, and telecommunications networks (Nickel et al., 2001; Alumur and Kara, 2009).

The design of an incomplete hub network is addressed on hub network design problems (HNDPs), which determine the location of hubs and select which hub arcs will be installed. As highlighted by Alumur et al. (2021), solving HNDPs is much more challenging than solving HLPs since HNDPs must handle the selection of hub arcs, in addition to locating hubs. Moreover, most HNDPs handle non-trivial routing of demands from origins to destinations since transportation routes can be quite varied, unlike HLPs, where the routes can have only one or two hubs. These features make the development of efficient solution methods for HNDP even more challenging.

This paper addresses a profit-oriented uncapacitated HNDP, with no special topological structure being imposed. The profit-maximizing multiple allocation HNDP (PMAHNDP) addressed here, proposed by Taherkhani and Alumur (2018), aims to determine the quantity and location of the hubs, allocate demand nodes to these hubs, establish the network topology, and route the demand flows to satisfy the demands between the origin and destination pairs that were selected to be served in order to obtain the highest profit. It also considers the multiple allocation strategy, does not allow direct connections between non-hub nodes, i.e., all demands must be routed through hubs, assumes that there are fixed costs involved in the installation of hubs and connections between them, and does not impose capacity constraints, so the hubs and hub arcs are assumed to have sufficient capacity to handle the demand flow routed in the network.

Potential applications of the PMAHNDP arise mainly in the design of transportation networks, in postal delivery and express delivery services. They can also be applied in any other sector that uses the structure of a hub network, in which profit is a determining factor.

The main contributions of this paper are:

- (i) We present an alternative formulation for the PMAHNDP, for which the only known formulation was the one proposed by Taherkhani and Alumur (2018). We also show that our formulation, which is based on the combination of other classical formulations in hub literature, is stronger, producing better linear programming relaxation bounds, than the formulation proposed by Taherkhani and Alumur (2018);
- (ii) We proposed two algorithms based on Benders decomposition methods. The method is improved through the inclusion of features such as the multiple cuts strategy, the insertion of non-dominated cuts (Pareto-optimal cuts), the use of an efficient algorithm to recover the subproblem objective function value, and a Benders-branch-and-cut procedure, in which cut filtering is performed. The efficiency of these proposed algorithms, a Benders decomposition algorithm and Benders-branch-and-cut algorithm, is evaluated through extensive computational experiments carried out with benchmark instances. Instances with up to 150 nodes have been solved. This number of nodes is the largest recorded in the literature, which we are aware of, for HNDP. As presented by Alumur et al. (2021), by reviewing studies addressing HNDPs based on cost minimization, only Contreras and Fernández (2014), which considering just trivial routing decisions, had solved instances with 125 nodes, while the other studies had solved only instances with up to 100 nodes. Furthermore, Taherkhani and Alumur (2018), which is the only study addressing the same problem addressed here, has solved only instances with up to 40 nodes by using the CPLEX solver. Hence, there is no solution procedure in the literature to solve large instances for this problem so far.

The rest of the paper is organized as follows. In Section 2, a review of the related literature is made. Section 3 presents the formal definition and a mixed integer programming (MIP) formulation for the problem. Section 4 contains the basic Benders decomposition algorithm, the Benders reformulation applied to our formulation, and a description of the techniques used to improve the convergence and efficiency of the method. A comparative analysis of our formulation with a formulation previously proposed to model the problem, as well as the description of the computational experiments, and an analysis of the results obtained are given in Section 5. Finally, in Section 6, final considerations are made.

2. Literature review

Different HNDPs can be found in the literature. Some of these studies address a specific topological structure, such as ring-star network (Lee and Ro, 1993), tree-star network (Contreras et al., 2010; Martins de Sá et al., 2013; Pessoa et al., 2016), star-star topology (Labbé and Yaman, 2008; Yaman and Elloumi, 2012) and hub lines network (Martins de Sá et al., 2015).

Other works addressed problems in which hubs can be partially interconnected, and no special topological restrictions are imposed on the network of hubs. Nickel et al. (2001) proposed a model for a HNDP applied to public transport networks, in which they considered the multiple allocation structure and allowed direct connections between non-hub nodes. Campbell et al. (2005a,b) addressed the hub arc location problems in which a fixed number of hub arcs is located to minimize total transport costs. Contreras and Fernández (2014) proposed an approach based on a general class of hub location and hub arc location problems focusing on minimizing a set of supermodular set functions. In particular, they dealt with the hub arc location problem, allowing only routes with at most one hub arc, solving AP instances with up to 125 nodes, and the supermodular problem, where instances with up to 50 nodes of AP data set were solved.

Yoon and Current (2008) studied a HNDP with the incorporation of fixed and variable costs for establishing hub arcs, considering multiple

allocation and direct connections between non-hub nodes. They used dual-based heuristics to solve instances with up to 100 nodes. The works of Alumur and Kara (2009) and Calik et al. (2009) considered the hub covering network design where the allocation is unique and the travel time between any pair of origin and destination nodes is within a given time limit. Davari et al. (2013) also studied the incomplete hub-covering location problem with the incorporation of uncertainty about the demand between the pairs of origin and destination nodes. Alumur et al. (2009) explored several versions of the single allocation HLP with incomplete hub network. The authors showed that, in some cases, an incomplete hub network could provide a service nearby to the service provided by a complete hub network, without affording the cost of installing all hub arcs.

Camargo et al. (2017) proposed mathematical formulations and Benders decomposition-based procedures for the HNPD with and without hop-constraints, which solved instances with up to 100 nodes. Xu et al. (2017) also presented a Benders decomposition algorithm to deal with the capacitated HNPD. Martins de Sá et al. (2018) proposed a Benders decomposition framework for solving a robust multiple allocation hub problem, in which direct connections between non-hub nodes were allowed, assuming uncertainties in demand flows and in the fixed costs for the location of the hubs. All the studies previously mentioned have dealt with cost minimization or hub covering problems.

There are few studies on HLPs and HNPDs addressing profit maximization. Alibeyg et al. (2016) is the first work on hub location from a profit perspective. This study proposes different versions of this problem by including additional restrictions or decisions, such as the requirement to meet the demand of all nodes in the network. It also considered the multiple allocation strategy, but despite dealing with a HNPD, it assumed that the service route of an origin and destination pair should contain at least one and at most two hub nodes. The performance of the models was evaluated using CPLEX with the CAB data set. In a subsequent study, Alibeyg et al. (2017) proposed an exact algorithm for solving the models presented in Alibeyg et al. (2016). The authors used reduction and partial enumeration tests to reduce the size of the problem and a branch-and-bound algorithm integrated with a Lagrangian relaxation. The CAB data set was also used to evaluate the performance of the proposed algorithm, and instances with up to 100 nodes were solved.

Taherkhani and Alumur (2018) proposed new models for the HNPD maximizing profit. The authors addressed all types of allocation strategies (single, multiple, and r -allocation), considering the cases where direct connections between non-hub nodes were allowed and were not allowed. The sets of CAB and AP instances were used to evaluate the performance of the proposed models using the CPLEX solver. Instances with up to 40 nodes were solved. Lin and Lee (2018) addressed an HLP with a profit-oriented objective in the design of an LTL cargo transportation network with time requirements. The article showed that the hub network generated by maximizing profit is denser than the network obtained by minimizing costs.

Taherkhani et al. (2020) addressed a profit maximizing capacitated multiple allocation HLP. The authors presented a deterministic model and a stochastic model, in which the demand is segmented from different classes; direct connections between non-hub nodes are not allowed; there are at most two hubs in the path of an origin and destination pair; the hub-level network is complete. For solving these models, the authors presented efficient solution methods based on Benders decomposition, in which reduction tests, having variables prefixed, were applied to reduce the size of the problem, as well as the computational effort. Instances with up to 500 nodes were solved using the deterministic model. However, as is already known in the HLP literature, problems with a complete hub-level network are less hard to solve than problems with incomplete hub networks (O'Kelly and Miller, 1994).

Other studies in the area of hub location focusing on profit maximizing have addressed problems that consider a competitive environment,

as Neamatian Monemi et al. (2017), Mahmoodjanloo et al. (2020), and Tiwari et al. (2021) or incorporate pricing decisions, like Lüer-Villagra and Marianov (2013), Čvokić and Stanimirović (2020), Esmaeili and Sedehzade (2020), and Sadeghi Dastaki et al. (2021). Furthermore, there are also some works, such as Tikani et al. (2018), Huo et al. (2019), and Salehi and Tikani (2020), that focused on determining the hub network in order to generate profit, integrating revenue management techniques in the problems.

We emphasize that the HNPD addressed in this article, described in detail in Section 3.1, differs from all the works discussed above considering profit maximizing, except Taherkhani and Alumur (2018), since no restrictions are imposed on the configuration of the hub network neither on the number of hub arcs used for routing demands. As previously mentioned in the Introduction Section, these two characteristics make the problem harder to solve.

3. Definitions and formulations of the problem

This section presents the general definitions and formulations to the profit maximizing multiple allocation HNPD (PMAHNPD). Section 3.1 introduces the general description of this problem. Section 3.2 shows a MIP formulation for it, based on formulations proposed by Campbell (1992), Ernst and Krishnamoorthy (1998) and Martins de Sá et al. (2018). Section 3.3 presents an alternative MIP formulation for the PMAHNPD and Section 3.4 discuss the properties of the two MIP formulations.

3.1. General description

The HLP studied in this paper determines the location of the hubs, the design of the hub network, and the routing of the demand flow that maximizes the total profit. The total profit is given by subtracting the total cost from the total revenue. The total revenue is obtained considering the amount of demand for each origin and destination pair served. The price is assumed to be an exogenous variable, determined by the market. It is also assumed that the revenue does not depend on the route of the demand flow. The total cost is composed of the total transportation cost and the fixed cost of installing the hubs, and the connections between them. The transportation cost for routing the demand flow between each origin and destination pairs is the sum of (i) the cost for transferring the demand from the origin to the first hub in the path (collection); (ii) the transportation cost in the hub arcs (transfer); and (iii) the transportation cost from the last hub in the path to the destination (distribution).

When computing the total transportation cost, it is assumed that there is a transportation cost economy of scale when routing the flow in the hub arcs. Economies of scale in the connections between the hub nodes arise, for instance, due to the consolidation of demand or the higher frequency of service in these facilities. This economy of scale is reflected in the transportation cost in the hub arcs through a constant flow independent discount factor.

The profit maximizing multiple allocation HNPD (PMAHNPD) addressed in this paper has the following characteristics:

- (i) a non-hub node can be allocated to as many hubs as needed;
- (ii) the number of hubs and hub arcs is not fixed beforehand;
- (iii) the hubs and the hub arcs in the network have sufficient capacity to handle all the necessary flow;
- (iv) it is not imposed that the hubs are all interconnected;
- (v) no particular structure of the hub network is assumed;
- (vi) there are fixed costs for installing hubs and also fixed costs for installing hub arcs;
- (vii) direct connections between non-hub nodes are not allowed; and
- (viii) it is not required that all demand be served.

Thus, the problem treated here is an uncapacitated multiple allocation HLP, seeking maximizing the profit in an incomplete network, which has fixed costs and does not allow direct connections between non-hub nodes. The formulation for the PMMAHNDP requires the definitions presented following.

Let N be the set of nodes that exchange flows in the network. This set of nodes is also the set of potential candidates to be a hub. The amount of demand to be sent from node $i \in N$ to node $j \in N$ is denoted by w_{ij} . For all $i, j \in N$, r_{ij} represents the revenue for serving a unit of demand from the origin node i to the destination node j . For all $i, j \in N$, c_{ij} represents the cost for routing a unit of demand through the arc connecting the nodes i and j . When the transportation is carried out through hub arcs, a constant discount factor given by α is applied, for $0 \leq \alpha < 1$. Hence, the transportation cost for each unit of demand routed through the hub arc connecting hubs k and m is equal to αc_{km} . The fixed cost of installing a hub on the $k \in N$ node is given by s_k and the fixed cost of operating the hub arc connecting the hub $k \in N$ and the hub $m \in N$ is denoted by g_{km} .

3.2. MIP formulation for the PMMAHNDP

This section concerns a MIP formulation for the PMMAHNDP. The proposed formulation combines the use of the four index variables from the formulation of [Campbell \(1992\)](#), for representing the flow on hub arcs, with the three index flow variables from the formulation of [Ernst and Krishnamoorthy \(1998\)](#), for representing flow in access arcs.

The binary location variable h_k indicates if a hub is located at the node $k \in N$ or not. The binary hub arc variable z_{km} indicates if a connection between the hubs $k \in N$ and $m \in N$ is active or not. The continuous flow variables $a_{ijk} \geq 0$ and $b_{ijm} \geq 0$ represent the fraction of the demand w_{ij} that enters and leaves the hub-level network through the hubs $k \in N$ and $m \in N$, respectively. The continuous flow variable $x_{ijkm} \geq 0$ determines the fraction of the demand w_{ij} that is routed in the arc between the hubs $k \in N$ and $m \in N$. A model for the PMMAHNDP, which will be denoted by (MOD1), is given by:

$$\begin{aligned}
 \text{(MOD1) max} \quad & \sum_{i \in N} \sum_{j \in N} \sum_{k \in N} r_{ij} w_{ij} a_{ijk} \\
 & - \left[\sum_{i \in N} \sum_{j \in N} \sum_{k \in N} c_{ik} w_{ij} a_{ijk} + \sum_{i \in N} \sum_{j \in N} \sum_{m \in N} c_{mj} w_{ij} b_{ijm} \right. \\
 & \quad + \sum_{i \in N} \sum_{j \in N} \sum_{k \in N} \sum_{m \in N} \alpha c_{km} w_{ij} x_{ijkm} \\
 & \quad \left. + \sum_{k \in N} s_k h_k + \sum_{k \in N} \sum_{m \in N} g_{km} z_{km} \right] \quad (1) \\
 \text{s. t.} \quad & \sum_{k \in N} a_{ijk} \leq 1 \quad i, j \in N \quad (2) \\
 & \sum_{m \in N} b_{ijm} \leq 1 \quad i, j \in N \quad (3) \\
 & a_{ijk} + \sum_{m \in N} x_{ijmk} = b_{ijk} + \sum_{m \in N} x_{ijkm} \quad i, j, k \in N \quad (4) \\
 & a_{ijk} \leq h_k \quad i, j, k \in N \quad (5) \\
 & b_{ijm} \leq h_m \quad i, j, m \in N \quad (6) \\
 & x_{ijkm} \leq z_{km} \quad i, j, k, m \in N, k \neq m \quad (7) \\
 & z_{km} \leq h_k \quad k, m \in N, k \neq m \quad (8) \\
 & z_{km} \leq h_m \quad k, m \in N, k \neq m \quad (9) \\
 & x_{ijkm} \geq 0 \quad i, j, k, m \in N, k \neq m \quad (10) \\
 & a_{ijk} \geq 0 \quad i, j, k \in N \quad (11) \\
 & b_{ijm} \geq 0 \quad i, j, m \in N \quad (12) \\
 & h_k \in \{0, 1\} \quad k \in N \quad (13) \\
 & z_{km} \in \{0, 1\} \quad k, m \in N, k \neq m. \quad (14)
 \end{aligned}$$

The objective function (1) represents the total net profit, which is given by the difference between the total revenue and the total cost. The first term of the objective function computes the revenue obtained from meeting the demand. The first three terms in parentheses represent transport costs, composed of access, distribution and transfer costs on the network. The last two terms in the parentheses correspond to the fixed costs of installing hubs and hub arcs, respectively. The sets of constraints (2) and (3) indicate that the total fraction of the demand for each pair of origin and destination nodes routed on the network is at most one. Flow conservation constraints (4) ensure that all demands that enter a given hub node must leave it. Constraints (5) and (6) ensure that the demand between the origin and destination pairs can be routed only through installed hubs. The set of constraints (7) ensures that the demand flow can be routed only through installed hub arcs. Constraints (8) and (9) indicate that a hub arc can only be operated if its two end hub nodes were installed. Constraints (10)–(14) represent the domain of the non-negative and binary variables.

3.3. Formulation of [Taherkhani and Alumur \(2018\)](#)

The formulation (MOD1) will be compared with the formulation for the same problem proposed by [Taherkhani and Alumur \(2018\)](#), i.e., the formulation that considers multiple allocation and does not allow direct connections between non-hub nodes. To present this model it is necessary to define two new sets of variables: (i) the binary variables y_{ijkm} , which indicates if the demand between nodes $i \in N$ and $j \in N$ is served along a path in which $k \in N$ is the first hub and $m \in N$ is the last hub; (ii) the flow variables f_{ikm} , which represent the amount of demand originating in node $i \in N$ that is routed in the hub arc between the hubs $k \in N$ and $m \in N$. The mathematical formulation from [Taherkhani and Alumur \(2018\)](#), denoted as (MOD2), is given by:

$$\begin{aligned}
 \text{(MOD2) max} \quad & \sum_{i \in N} \sum_{j \in N} \sum_{k \in N} \sum_{m \in N} r_{ij} w_{ij} y_{ijkm} - \left[\sum_{i \in N} \sum_{j \in N} \sum_{k \in N} \sum_{m \in N} (c_{ik} + c_{mj}) w_{ij} y_{ijkm} \right. \\
 & \quad \left. + \sum_{i \in N} \sum_{k \in N} \sum_{m \in N} \alpha c_{km} f_{ikm} + \sum_{k \in N} s_k h_k + \sum_{k \in N} \sum_{m \in N} g_{km} z_{km} \right] \quad (15) \\
 \text{s. t.} \quad & (8), (9), (13), (14), \quad (16) \\
 & \sum_{k \in N} \sum_{m \in N} y_{ijkm} \leq 1 \quad i, j \in N \quad (17) \\
 & \sum_{m \in N} y_{ijkm} + \sum_{m \in N} y_{ijmk} \leq h_k \quad i, j, k \in N \quad (18) \\
 & \sum_{m \in N} f_{imk} + \sum_{j \in N} \sum_{m \in N} w_{ij} y_{ijkm} = \sum_{m \in N} f_{ikm} + \sum_{j \in N} \sum_{m \in N} w_{ij} y_{ijmk} \quad i, k \in N \quad (19) \\
 & f_{ikm} \leq \sum_{j \in N} w_{ij} z_{km} \quad i, k, m \in N, k \neq m \quad (20) \\
 & f_{ikm} \geq 0 \quad i, k, m \in N \quad (21) \\
 & y_{ijkm} \in \{0, 1\} \quad i, j, k, m \in N. \quad (22)
 \end{aligned}$$

The objective function (15) maximizes the total net profit similarly to the objective function of (MOD1): the first term calculates the revenue obtained by serving the demand of the selected node pairs and the terms in parentheses represent the total network design costs, composed of transportation costs (the first two terms) and the fixed costs of installing hubs and hub arcs (the last two terms). Constraints (17) ensure that if the demand between a given pair of nodes is met, only one pair of hubs will be selected to be the first and the last hubs in the path. Constraints (18) guarantee that the demand between the origin and destination nodes can be satisfied only through installed hubs. Constraints (19) are flow conservation constraints. The set of constraints (20) enforces that the demand flow is routed only through active hub arcs. Constraints (21) and (22) represent the domain of the non-negative and binary variables.

According to [Taherkhani and Alumur \(2018\)](#), as the problem is uncapacitated, then the demand between each pair of origin and destination is completely satisfied or not. Thus, even if the integrality of

Table 1

Number of variables and constraints of our formulation (MOD1) and of the formulation (MOD2) of multiple allocation from Taherkhani and Alumur (2018) depending on the size of the instance.

Properties	Formulation (MOD1)	Formulation (MOD2)
Binary variables	n^2	$n^4 + n^2$
Total of variables	$n^4 + n^3 + n^2$	$n^4 + n^3 + n^2$
Total of constraints	$n^4 + 2n^3 + 4n^2 - 2n$	$2n^3 + 3n^2 - 2n$

the variables y_{ijkm} is relaxed, there is always an optimal solution in which $y_{ijkm} \in \{0, 1\}$, $\forall i, j, k, m \in N$. In the experiments carried out whose results are presented in Section 5.1.2, this variable was relaxed to allow the comparison of the application of Benders decomposition algorithm of the CPLEX solver in both formulations.

Note that the (MOD1) and (MOD2) formulations, presented for the profit-oriented objective HNDP, have a basic structure similar to the cost-oriented objective problems. However, as discussed earlier, the problem addressed here is more complex than classic HLPs and HNDPs, because it has an extra level of decision. The main differences between profit-oriented and cost-oriented objective models are, as expected, in the objective of the problem. In the expression of the objective function, we compute the profit, while cost-oriented models calculate the costs associated with the hub network design. Furthermore, in cost-oriented problems, the constraints (2), (3) and (17) would be equality constraints, ensuring that all pairs of nodes would be met, without having to decide which ones of them would be served.

3.4. Formulations properties

The formulations (MOD1) and (MOD2), although representing the same problem, have different characteristics. The main difference between these two formulations refers to how the network flow is modeled in each of them. The formulations also differ regarding the number of binary variables and the number of constraints in each one. Table 1 presents a summary of these data. According to this table, the two formulations have the same number of variables. However, the number of binary variables in formulation (MOD1) is less than the number of binary variables in formulation (MOD2). On the other hand, our formulation involves more constraints than the formulation (MOD2) from Taherkhani and Alumur (2018).

To deepen the comparison between (MOD1) and (MOD2) formulations, consider the characterization of the stronger formulation introduced by Bertsimas and Tsitsiklis (1997), showed in Definition 1.

Definition 1 (Bertsimas and Tsitsiklis, 1997). Let F_1 and F_2 be formulations for the same MIP problem and consider P_1 and P_2 the corresponding feasible regions of their linear programming relaxations. The formulation F_1 is said to be stronger than the formulation F_2 if P_1 is strictly contained in P_2 .

For a maximization problem, saying that a formulation F_1 is stronger than a formulation F_2 means that the upper bound obtained by the LP relaxation of F_1 is closer to the objective function value of the optimal solution of the MIP than the value obtained by the linear programming relaxation of F_2 or, then, both bounds are equal. Proposition 1 shows that our formulation (MOD1) is stronger than the formulation (MOD2) proposed by Taherkhani and Alumur (2018).

Proposition 1. *The formulation (MOD1) is stronger than the formulation (MOD2).*

Proof. The LP relaxations of the formulations (MOD1) and (MOD2) will be denoted by $\bar{F}_1$ and $\bar{F}_2$, respectively. Let P_1 and P_2 be the feasible regions of $\bar{F}_1$ and $\bar{F}_2$, respectively. It will be proved that $P_1 \subset P_2$. As there are distinct variables between the formulations, we will show

that, given any solution in P_1 , there is an equivalent solution to it in P_2 , i.e., a feasible solution that has the same objective function value for the problem. Let $s_1 = (h, z, a, b, x)$ be a solution in P_1 , where each component represents a vector corresponding to the variables h_k , z_{km} , a_{ijk} , b_{ijm} and x_{ijkm} . A solution $s_2 \in P_2$ with the same objective value of s_1 can be constructed as follows. Keep the values of the variables h and z . Let R denotes the set of pairs (k, m) such that k and m are, respectively, the first and last hub of a route used to serve the pair (i, j) ; let R^1 be composed by the hubs k such that k is the first hub of a route used to attend the demand of (i, j) , and let R^2 be the set of hubs m such that m is the last hub of a route used to attend the demand of the pair (i, j) . Then, for each pair $(i, j) \in N \times N$ whose demand was satisfied, find the values of the variables y by solving the system of equations:

$$\sum_{(k,m) \in R} y_{ijkm} = \sum_{k: (k,m) \in R} a_{ijk} \quad (23)$$

$$\sum_{m: (k,m) \in R} y_{ijkm} = a_{ijk}, \quad \forall k \in R^1 \quad (24)$$

$$\sum_{k: (k,m) \in R} y_{ijkm} = b_{ijm}, \quad \forall m \in R^2 \quad (25)$$

Note that this system of equations determines the fraction of demand w_{ij} that is routed along each of the paths connecting origin node i to destination node j . Due to the flow balancing constraint (4), which guarantee that $\sum_{k \in N} a_{ijk} = \sum_{m \in N} b_{ijm}$, then this system is always feasible. Furthermore, the value of the other variables y_{ijkm} that is not in this system is set as zero. In the sequel, for each $i, k, m \in N$, define the variables f as:

$$f_{ikm} = \sum_{j \in N} w_{ij} x_{ijkm}. \quad (26)$$

Proposition 2 in Appendix A shows that the solution $s_2 = (h, z, y, f)$, defined as above, belongs to P_2 and produces the same objective function value as s_1 . To verify that P_1 and P_2 are distinct feasible regions, consider $\bar{s}_2$ a solution for P_2 , so that there is a hub arc (k, m) installed such that, for all node $i \in N$,

$$f_{ikm} < \sum_{j \in N} w_{ij}, \quad (27)$$

and that satisfies the condition that all demand from node i directed to some node $j \in N$ is routed through the hub arc (k, m) . Thus, in any solution of P_1 with the same routing decision of $\bar{s}_2$, to satisfy the set of constraints (7), the value of z_{km} must be necessarily equal to one, since $x_{ijkm} = 1$ for some pair $(i, j) \in N \times N$. On the other hand, in P_2 , the value of z_{km} can be strictly less than one (See constraints (20)). This shows that $P_1 \neq P_2$.

Since we are dealing with a maximization problem and this variable affects the costs of the network design, in the optimal solutions for $\bar{F}_2$ the value of this variable will be

$$z_{km} = \max_{i \in N} \left\{ \frac{f_{ikm}}{\sum_{j \in N} w_{ij}} \right\}. \quad (28)$$

Thus, in this case, the value of the total cost of the network design given by a solution of $\bar{F}_2$ that satisfies the condition (27), for some installed hub arc (k, m) and $i \in N$, will be strictly less than the value obtained with a solution of $\bar{F}_1$, with equivalent flow routing, where $x_{ijkm} = 1$, for some $(i, j) \in N \times N$. Consequently, $\bar{F}_2$ will give a solution that has an objective function value strictly less than the value of a solution of $\bar{F}_1$. $\square$

Therefore, we choose the formulation (MOD1) to apply the Benders decomposition method since it is known that this method has a better performance when applied to strong formulations (Magnanti and Wong, 1981).

4. Benders decomposition

The Benders decomposition method is a partitioning procedure, proposed by Benders (1962), to solve optimization problems with mixed variables, such as the PMMAHNDP, which is a mixed-integer linear problem. The method is based on the decomposition of the original problem into two problems, known as the master problem (MP) and the Benders subproblem (SP). The MP is a reformulation of the original problem that contains only a subset of its original variables and their associated constraints, besides an additional set of constraints, known as Benders cuts (BCs). The Benders subproblem (SP) corresponds to the original problem with the MP variables temporarily fixed. For a mixed-integer linear problem, it is common, in the literature, to keep integer variables in the MP, and to keep the other variables in the SP.

The Benders cuts (BCs) of the MP are originated from extreme points and extreme rays of the polyhedron formed by the constraints of the dual problem of the Benders subproblem (DSP). The expression referring to BCs is given by the expression of the DSP objective function, where the dual variables are replaced by the values of the extreme points or extreme rays, and the master problem variables, which were temporarily fixed in the SP and DSP, are variable in the BCs. Since, in general, the number of BCs grows exponentially as the instance size increases, and most of them are not active in the optimal solution, most BCs are initially relaxed in the first iterations of the Benders decomposition method.

Therefore, during the execution of the method, the relaxed MP and the SP are solved iteratively, so that, in each iteration, a new BC is added to the MP. This BC is found from the optimal solution of the DSP, with the optimal solution of the current MP fixed. This cycle is repeated until the convergence of the upper bound and the lower bound is attained, if there is an optimal solution to the original problem.

As the problem addressed in this work has a maximization objective, and the MP is a relaxation of the original problem, so it provides an upper bound (UB) for this problem. On the other hand, a lower bound (LB) can be obtained by combining part of the objective function of the master problem with the solution of the SP. Rahmaniani et al. (2017) presented a review of the literature concerning the Benders decomposition method.

The Benders decomposition method has been successfully applied in many studies to solve different variants of the HLP, including uncapacitated multiple allocation problem (Camargo et al., 2008; Contreras et al., 2011; O'Kelly et al., 2015; Ghaffarinasab, 2020) and capacitated HLPs (Contreras et al., 2012; Merakli and Yaman, 2017), single allocation problem under congestion (Camargo and Miranda, 2012; Najy and Diabat, 2020), problems applied to transportation networks (Gelareh and Nickel, 2011), and incomplete hub networks problems (Martins de Sá et al., 2013, 2015; Camargo et al., 2017; Xu et al., 2017; Martins de Sá et al., 2018).

The main difference between the current work and the other studies that applied the Benders decomposition method to similar HNDP is: (i) this paper addresses a different HNDP with a different objective function and constraints, and (ii) the most successful variant of the Benders decomposition in solving instances of the PMMAHNDP is composed of a different combination of tools to improve the convergence of the method.

The following section presents the reformulation of Benders decomposition method using our formulation and the strategies adopted to improve the performance of the method.

4.1. Benders reformulation

This section presents a Benders reformulation for the model (MOD1). Consider that the variables regarding the location of the hubs and the activation of the hub arcs are put in the master problem (MP), and the other decision variables are in the primal subproblem (SP).

Then, fixing the vectors of integer variables h and z with values h^e and z^e , respectively, we get the linear primal subproblem (SP):

$$(SP) \quad \max \quad \sum_{i \in N} \sum_{j \in N} \sum_{k \in N} (r_{ij} - c_{ik}) w_{ij} a_{ijk} - \left[\sum_{i \in N} \sum_{j \in N} \sum_{m \in N} c_{mj} w_{ij} b_{ijm} + \sum_{i \in N} \sum_{j \in N} \sum_{k \in N} \sum_{m \in N} \alpha c_{km} w_{ij} x_{ijkm} \right] \quad (29)$$

$$\text{s.t.} \quad \sum_{k \in N} a_{ijk} \leq 1 \quad i, j \in N \quad (30)$$

$$\sum_{m \in N} b_{ijm} \leq 1 \quad i, j \in N \quad (31)$$

$$a_{ijk} - b_{ijk} + \sum_{m \in N} x_{ijmk} - \sum_{m \in N} x_{ijkm} = 0 \quad i, j, k \in N \quad (32)$$

$$a_{ijk} \leq h_k^e \quad i, j, k \in N \quad (33)$$

$$b_{ijm} \leq h_m^e \quad i, j, m \in N \quad (34)$$

$$x_{ijkm} \leq z_{km}^e \quad i, j, k, m \in N, k \neq m \quad (35)$$

$$x_{ijkm} \geq 0 \quad i, j, k, m \in N, k \neq m \quad (36)$$

$$a_{ijk} \geq 0 \quad i, j, k \in N \quad (37)$$

$$b_{ijm} \geq 0 \quad i, j, m \in N \quad (38)$$

Associating the dual variables $\beta_{ij} \geq 0$, $\gamma_{ij} \geq 0$, $\delta_{ijk} \in \mathbb{R}$, $\theta_{ijk} \geq 0$, $\mu_{ijm} \geq 0$ and $\rho_{ijkm} \geq 0$ with the constraints (30)–(35), respectively, the dual subproblem (DSP) can be written as:

$$(DSP) \quad \min \quad \sum_{i \in N} \sum_{j \in N} (\beta_{ij} + \gamma_{ij}) + \sum_{i \in N} \sum_{j \in N} \sum_{k \in N} h_k^e \theta_{ijk} + \sum_{i \in N} \sum_{j \in N} \sum_{m \in N} h_m^e \mu_{ijm} + \sum_{i \in N} \sum_{j \in N} \sum_{k \in N} \sum_{m \in N} z_{km}^e \rho_{ijkm} \quad (39)$$

$$\text{s.t.} \quad \beta_{ij} + \delta_{ijk} + \theta_{ijk} \geq w_{ij}(r_{ij} - c_{ik}) \quad i, j, k \in N \quad (40)$$

$$\gamma_{ij} - \delta_{ijm} + \mu_{ijm} \geq -w_{ij}c_{mj} \quad i, j, m \in N \quad (41)$$

$$\delta_{ijm} - \delta_{ijk} + \rho_{ijkm} \geq -\alpha w_{ij}c_{km} \quad i, j, k, m \in N, k \neq m \quad (42)$$

$$\beta_{ij}, \gamma_{ij} \geq 0 \quad i, j \in N \quad (43)$$

$$\theta_{ijk} \geq 0 \quad i, j, k \in N \quad (44)$$

$$\mu_{ijm} \geq 0 \quad i, j, m \in N \quad (45)$$

$$\rho_{ijkm} \geq 0 \quad i, j, k, m \in N, k \neq m \quad (46)$$

$$\delta_{ijk} \in \mathbb{R} \quad i, j, k \in N \quad (47)$$

Observe that the primal subproblem (SP), given by Expressions (29)–(38), is always feasible and bounded, since meeting the demand of the origin and destination pairs is not mandatory, being carried out only when it is profitable, and there is a finite number of demand pairs to be served. From the Strong Duality Theorem (Bertsimas and Tsitsiklis, 1997), it follows that the dual subproblem (DSP), given by Expressions (39)–(47), is always feasible, and the optimal solution value of the (DSP) is also bounded.

Let $\mathbb{V}$ denote the set of extreme points of the polyhedron formed from constraints (40)–(47), and p^v a generic extreme point such that $p^v \in \mathbb{V}$ and $p^v = (\beta^v, \gamma^v, \theta^v, \mu^v, \rho^v, \delta^v)$. Then, the Benders cuts (BC) to be added to the master problem (MP) can be given as:

$$\eta \leq \sum_{i \in N} \sum_{j \in N} \left[\beta_{ij}^v + \gamma_{ij}^v + \sum_{k \in N} h_k \theta_{ijk}^v + \sum_{m \in N} h_m \mu_{ijm}^v + \sum_{k \in N} \sum_{m \in N, m \neq k} z_{km} \rho_{ijkm}^v \right], \quad \forall (\beta^v, \gamma^v, \theta^v, \mu^v, \rho^v, \delta^v) \in \mathbb{V}, \quad (48)$$

where η is the master problem variable that overestimates the total profit regarding the revenues and transportation costs. Consequently, the (MP) is given by:

$$(MP) \quad \max \quad \eta - \sum_{k \in N} s_k h_k - \sum_{k \in N} \sum_{m \in N, m \neq k} g_{km} z_{km} \quad (49)$$

$$\text{s.t.} \quad (8), (9), (13), (14) \quad (50)$$

$$\eta \leq \sum_{i \in N} \sum_{j \in N} (\beta_{ij}^v + \gamma_{ij}^v) + \sum_{i \in N} \sum_{j \in N} \sum_{k \in N} h_k \theta_{ijk}^v + \sum_{i \in N} \sum_{j \in N} \sum_{m \in N} h_m \mu_{ijm}^v + \sum_{i \in N} \sum_{j \in N} \sum_{k \in N} \sum_{m \in N} z_{km} \rho_{ijkm}^v \quad (\beta^v, \gamma^v, \theta^v, \mu^v, \rho^v, \delta^v) \in \mathbb{V} \quad (51)$$

$$0 \leq \eta \leq M \quad (52)$$

where M is a valid upper bound for the value of the variable η . The constraint (52) ensures that the (MP) is not unbounded in the initial iteration of the Benders decomposition method when no BCs had been added.

An outline of the classical Benders decomposition method, applied to the PMMAHNDP, is described in Algorithm 1. In this algorithm, $v(MP)$ and $v(DSP)$ denote the value of the current optimal solution of the (MP) and the (DSP), respectively; UB and LB are the current upper and lower bounds, respectively; ε is the user-defined optimality gap tolerance. In addition, (h^e, z^e) and $(\beta^e, \gamma^e, \theta^e, \mu^e, \rho^e, \delta^e)$ are the solutions obtained from the (MP) and the (DSP) in the iteration e , respectively, and $\mathbb{V}^e \subset \mathbb{V}$ is the set of extreme points generated up to the iteration e and used to compose the current set of BCs.

Algorithm 1: Classical Benders Decomposition

```

1 Let  $UB = +\infty, LB = -\infty, gap = +\infty, e = 0, \mathbb{V}^e = \emptyset$ 
2 while ( $gap > \varepsilon$ ) do
3   Solve the MP (49)–(52) and obtain  $(h^e, z^e)$  and  $v(MP)$ 
4    $UB \leftarrow v(MP)$ 
5   Solve the DSP (39)–(47) and obtain  $(\beta^e, \gamma^e, \theta^e, \mu^e, \rho^e, \delta^e)$  and  $v(DSP)$ 
6    $\mathbb{V}^{e+1} \leftarrow \mathbb{V}^e \cup \{(\beta^e, \gamma^e, \theta^e, \mu^e, \rho^e, \delta^e)\}$ 
7    $LB \leftarrow \max \left\{ LB, v(DSP) - \sum_{k \in N} s_k h_k^e - \sum_{k \in N} \sum_{m \in N, m \neq k} g_{km} z_{km}^e \right\}$ 
8    $gap \leftarrow (UB - LB) / UB$ 
9    $e \leftarrow e + 1$ 
10 end
```

In the following section, some strategies to improve this classic version of the Benders decomposition method are presented.

4.2. Benders decomposition enhancements

The computational efficiency of the Benders decomposition algorithm is, in general, related to the chosen mathematical formulation, the computational effort required to solve the (MP) and the (DSP), and the quality of the Benders cuts generated at each iteration. Many techniques to improve the performance of the method have been proposed in the literature.

Geoffrion and Graves (1974) proposed to generate BCs without solving the MP to optimality, using a tolerance factor. McDaniel and Devine (1977) proposed a relaxation of the MP's integrality constraints during the first iterations, allowing also to generate cuts without solving an integer program. More recently, Contreras et al. (2011) included tests to reduce the size of the problem in the internal iterations of the algorithm, as well as used non-dominated cuts. Taherkhani et al. (2020) also used reduction tests and variable pre-fixation to reduce the problem size and computational effort.

Magnanti and Wong (1981) proposed to generate Pareto-optimal cuts and add these cuts to the MP to improve the convergence of the method for cases in which the DSP is degenerate. Pareto-optimal cuts are those that are not dominated by any other cut. The authors formulated an additional subproblem that allows generating Pareto-optimal cuts using the notion of core points. Papadakos (2008) presented an improved method for generating Pareto-optimal cuts and proved that it is possible to produce this type of cut from points that are not

core points of the MP solutions space. Hence, the generation of non-dominated cuts, in general, has the advantage of generating stronger cuts, which can reduce the number of iterations required to reach the method convergence at the cost of solving an additional subproblem.

In problems that have a decomposable structure, one of the techniques used to reduce the number of iterations of the algorithm is the addition of multiple BCs to each iteration (Birge and Louveaux, 1988).

One strategy to improve the convergence of the method is to generate BCs within the branch-and-cut tree when solving the MP (Codato and Fischetti, 2006; Naoum-Sawaya and Elhedhli, 2013; Martins de Sá et al., 2015). This strategy is named as Benders branch-and-cut. As mentioned by Naoum-Sawaya and Elhedhli (2013), the BCs can be generated for any potential incumbent solution. In this case, the method converges in a single iteration, or the BCs can be generated from the fractional solutions attained through the branch-and-cut search.

In the current work, we employed three strategies to improve the computational performance of the method: (i) the multiple cuts strategy; (ii) the generation of non-dominated cuts; and (iii) the use of Benders branch-and-cut structure. The application of these strategies to the problem treated here is described in the following.

4.2.1. Multiple cuts strategy

The multiple cuts strategy consists of adding multiple BCs to the MP for each iteration. Since the primal subproblem SP can be partitioned into n^2 independent subproblems, two strategies can be considered. The first one is to add n cuts from Benders per iteration, that is, one cut for each origin node $i \in N$. Thus, for any extreme points $(\beta^v, \gamma^v, \theta^v, \mu^v, \rho^v, \delta^v) \in \mathbb{V}$, the following set of cuts can be defined:

$$\eta_i \leq \sum_{j \in N} \left[\beta_{ij}^v + \gamma_{ij}^v + \sum_{k \in N} h_k \theta_{ijk}^v + \sum_{m \in N} h_m \mu_{ijm}^v + \sum_{k \in N} \sum_{m \in N, m \neq k} z_{km} \rho_{ijkm}^v \right], \quad \forall i \in N. \quad (53)$$

In this case, the variable η in the objective function (49) of the MP is replaced by $\sum_{i \in N} \eta_i$.

The second strategy is to add n^2 BCs to the MP in each iteration, i.e., for any extreme point $(\beta^v, \gamma^v, \theta^v, \mu^v, \rho^v, \delta^v) \in \mathbb{V}$, the following set of BCs, one for each pair of origin and destination nodes (i, j) , can be defined:

$$\eta_{ij} \leq \beta_{ij}^v + \gamma_{ij}^v + \sum_{k \in N} h_k \theta_{ijk}^v + \sum_{m \in N} h_m \mu_{ijm}^v + \sum_{k \in N} \sum_{m \in N, m \neq k} z_{km} \rho_{ijkm}^v, \quad \forall i, j \in N. \quad (54)$$

The variable η in the MP objective function (49), in this case, is replaced by $\sum_{i \in N} \sum_{j \in N} \eta_{ij}$.

4.2.2. Inserting non-dominated cuts

This paper consider the procedure presented by Papadakos (2008) for the generation of Pareto-optimal cuts. As before, let $p^v = (\beta^v, \gamma^v, \theta^v, \mu^v, \rho^v, \delta^v) \in \mathbb{V}$ be a extreme point. In the context of the PMMAHNDP, if a Benders cut associated with the extreme point $p^{v_1} \in \mathbb{V}$ dominates the Benders cut corresponding to $p^{v_2} \in \mathbb{V}$, then:

$$\begin{aligned} \beta_{ij}^{v_1} + \gamma_{ij}^{v_1} + \sum_{k \in N} h_k \theta_{ijk}^{v_1} + \sum_{m \in N} h_m \mu_{ijm}^{v_1} \\ + \sum_{k \in N} \sum_{m \in N, m \neq k} z_{km} \rho_{ijkm}^{v_1} \leq \beta_{ij}^{v_2} + \gamma_{ij}^{v_2} + \sum_{k \in N} h_k \theta_{ijk}^{v_2} \\ + \sum_{m \in N} h_m \mu_{ijm}^{v_2} + \sum_{k \in N} \sum_{m \in N, m \neq k} z_{km} \rho_{ijkm}^{v_2}, \quad \forall (h, z) \in \mathbb{Y}, \end{aligned} \quad (55)$$

with strict inequality for at least one point $(h, z) \in \mathbb{Y}$, where $\mathbb{Y} = \{(h, z) : (8), (9), (13), (14)\}$.

In the procedure proposed by Papadakos (2008), Pareto-optimal cuts are obtained using an alternative subproblem, called the independent Magnanti–Wong subproblem. This subproblem is independent

Algorithm 2: A Benders decomposition based on Papadakos (2008)

```

1 Let  $UB = +\infty, LB = -\infty, gap = +\infty, e = 0, \mathbb{V}^e = \emptyset, \lambda = 0.5$ .
2 Find an initial Magnanti–Wong point  $(h^0, z^0)$ 
3 while ( $gap > \epsilon$ ) do
4   Solve the IMWSP (56)–(57) and obtain  $(\beta^e, \gamma^e, \theta^e, \mu^e, \rho^e, \delta^e)$ 
5    $\mathbb{V}^{e+1} \leftarrow \mathbb{V}^e \cup \{(\beta^e, \gamma^e, \theta^e, \mu^e, \rho^e, \delta^e)\}$ 
6   Solve the MP (49)–(52) and obtain  $(h^e, z^e)$  and  $v(MP)$ 
7    $UB \leftarrow v(MP)$ 
8   Solve the SP (29)–(38) and obtain  $v(SP)$ 
9    $LB \leftarrow \max \left\{ LB, v(SP) - \sum_{k \in N} s_k h_k^e - \sum_{k \in N} \sum_{m \in N, m \neq k} g_{km} z_{km}^e \right\}$ 
10  Update the Magnanti–Wong point using (58)–(59)
11   $gap \leftarrow (UB - LB)/UB$ 
12   $e \leftarrow e + 1$ 
13 end

```

of the DSP solution and requires only the use of Magnanti–Wong points, i.e., points that provide Pareto-optimal cuts when applied to the subproblems defined by Magnanti and Wong (1981) or by Papadakos (2008). The independent Magnanti–Wong subproblem (IMWSP) for the PMMAHNDP can be given as:

$$\min \quad \sum_{i \in N} \sum_{j \in N} (\beta_{ij} + \gamma_{ij}) + \sum_{i \in N} \sum_{j \in N} \sum_{k \in N} h_k^0 \theta_{ijk} + \sum_{i \in N} \sum_{j \in N} \sum_{m \in N} h_m^0 \mu_{ijm} \\ + \sum_{i \in N} \sum_{j \in N} \sum_{k \in N} \sum_{m \in N, m \neq k} z_{km}^0 \rho_{ijkm} \quad (56)$$

$$\text{s.t.} \quad (40)–(47), \quad (57)$$

where (h^0, z^0) is a Magnanti–Wong point.

For many problems, finding good *core points* or Magnanti–Wong points is not an easy task (Mercier et al., 2005). However, Papadakos (2008) showed that any convex combination of a Magnanti–Wong initial point and a point belonging to the MP solutions space is also a Magnanti–Wong point.

This study considers the point (h^0, z^0) given by $h_k^0 = 0.5, \forall k \in N$, and $z_{km}^0 = 0.4, \forall k, m \in N, k \neq m$, which is a core point of $\mathbb{V}$, as the starting Magnanti–Wong point. The Magnanti–Wong point update, at each iteration, is done by:

$$h_{e+1}^0 = (1 - \lambda)h_e^0 + \lambda h^e, \quad (58)$$

$$z_{e+1}^0 = (1 - \lambda)z_e^0 + \lambda z^e, \quad (59)$$

where (h^e, z^e) corresponds to the optimal solution of the MP in the iteration e and $0 < \lambda < 1$. Empirically, $\lambda = 0.5$ provides the best results (Papadakos, 2008; Mercier et al., 2005; Xu et al., 2017). Note that, since the DSP for the PMMAHNDP is always feasible, the update rule using (58)–(59) provides a valid Magnanti–Wong point for the problem in all iterations of the decomposition method Benders.

Algorithm 2 outlines the Benders decomposition algorithm for the PMMAHNDP with the Pareto-optimal cut generation scheme, where $v(SP)$ represents the optimal objective value for the SP. Note in the Algorithm 2 that only Pareto-optimal cuts are added to the MP through the solutions of the IMWSP (56)–(57) (lines 4–5 of Algorithm 2) and that the SP is solved exclusively to update the LB by finding the value for $v(SP)$ (line 8 of Algorithm 2). Therefore, to obtain the value of $v(SP)$ efficiently, it is used an adapted version of the well-known Floyd–Warshall algorithm (Floyd, 1962), which allows finding the shortest paths between the pair of nodes.

The value of $v(SP)$ can be determined as follows. Let C_{ij} be the minimum unit transportation cost from the origin $i \in N$ to the destination $j \in N$. Let P_{ij} be the unit profit associated with the pair

$i, j \in N$, which can be calculated as:

$$P_{ij} = \max\{0, r_{ij} - C_{ij}\}.$$

Thus $v(SP)$ can be given by:

$$v(SP) = \sum_{i \in N} \sum_{j \in N} w_{ij} P_{ij}.$$

Here, C_{ij} can be determined using the Floyd–Warshall algorithm as follows. Instead of applying the Floyd–Warshall algorithm to a network composed of all nodes, the algorithm is applied to determine the lowest unit transportation costs among only the pairs of hubs in the hub-level network, assuming that the transportation cost associated with non-active hub arcs is large enough for not being considered. Let $H^1 \subset N$ be the set of active hubs, and let C_{km}^{FW} be the lowest unit transportation cost between hubs $k, m \in H^1$. Then, for all $i, j \in N$, C_{ij} is given by:

$$C_{ij} = \min_{k, m \in H^1} \{c_{ik} + C_{km}^{FW} + c_{mj}\}.$$

It is important to highlight that the Floyd–Warshall algorithm is used only to evaluate a solution to the Benders SP (line 8 of Algorithm 2), which consists of finding the maximal revenue for an HNDP with a non-trivial route. Note that the evaluation of a solution for the Benders SP, i.e., the computation of $v(SP)$, could be achieved by solving the DSP using a solver, rather than using the Floyd–Warshall algorithm. However, solving the DSP without decomposition would be impractical for large instances. On the other hand, solving the DSP decomposed in smaller subproblems would demand more computational time, since the resolution of each subproblem should handle the non-trivial routes in the hub network, which is done only once in the proposed Floyd–Warshall algorithm. Unfortunately, as the HNDP addressed here allows non-trivial routes, it is challenging to develop a specialized algorithm to solve the IMWSP since it deals with fractional values for h_k^0 and z_{km}^0 . Therefore, even solving the SP with this method, the proposed formulation is used to define the IMWSP (56)–(57) that generate Pareto-optimal cuts by means of the CPLEX solver (line 4 of Algorithm 2).

4.2.3. Benders branch-and-cut scheme

The Benders branch-and-cut (BBC) scheme consists of adding Benders cuts (BCs) within a standard structure of the branch-and-cut algorithm. This procedure has been used in many studies addressing HLPs, in which the addition of cuts to the branch-and-cut tree is performed in several different ways.

Fortz and Poss (2009) and Gelareh and Nickel (2011) presented a BBC approach in which BCs are generated for each potential incumbent solution. Naoum-Sawaya and Elhedhli (2013), in turn, used a strategy of adding BCs in each node of the tree. Adulyasak et al. (2015) and Martins de Sá et al. (2015) proposed a BBC approach considering three cut generation strategies: (i) add BCs only for potential incumbent solutions; (ii) adding BCs for any potential incumbent solution and any fractional solution at the root node; and (iii) add BCs for any potential incumbent solution and any fractional solution. It is important to highlight that all strategies adding BCs for any potential incumbent solution result in a single iteration of the Benders decomposition method.

This paper uses a BBC scheme considering the three cut generation strategies (i), (ii), and (iii), adopted by Adulyasak et al. (2015) and Martins de Sá et al. (2015), and a fourth strategy, in which the BCs are added to the tree only from fractional solutions at the root node. This last strategy allows the insertion of cuts in the tree, and the insertion of BCs after each resolution of the MP, in the traditional iterations of the methods. In all these strategies, only Pareto-optimal cuts are added, according to what is presented in Section 4.2.2.

5. Computational experiments

This section presents the computational experiments performed to evaluate the methods presented in Section 4, as well as an analysis of the found results. Section 5.1 presents a comparison between

the proposed formulation (MOD1) and the formulation (MOD2) from [Taherkhani and Alumur \(2018\)](#). The performance of the proposed Benders decomposition approaches is evaluated in Section 5.2.

5.1. Comparison of formulations

The main purpose of the computational experiments presented in this section is to make a comparison of the formulation (MOD1) here proposed with the multiple allocation mathematical model shown in formulation (MOD2), proposed by [Taherkhani and Alumur \(2018\)](#).

The experiments were performed on an Intel Core i7-7500U computer, 2.70 GHz, with 16 GB of RAM, using the Linux environment. The model was coded in C++ using Concert Technology, and the CPLEX 12.10 solver was used to solve the problem with the default configuration. The CPLEX's Benders decomposition method was also employed to solve the instances. The MIP tolerance gap considered in the experiments was 10^{-7} .

To carry out the experiments, two benchmark data sets from the hub location area were used: the US Civil Aeronautics Board (CAB) data set ([O'Kelly, 1987](#)) and the Australian Post (AP) data set ([Ernst and Krishnamoorthy, 1996](#)). Section 5.1.1 details the prefixing of variables applied in the computational experiments. In Section 5.1.2, the results of the tests with the CAB data set are presented, while the results with the AP data set are shown in Section 5.1.3.

5.1.1. Pre-fixation and relaxation of variables

As discussed earlier, the PMMAHNDP has the property that the demand between a pair of origin and destination nodes is only satisfied if it is profitable. This property was used to pre-fix the value of some variables to decrease the computational effort of the formulations in the experiments carried out in this section. Thus, we assume that, in our formulation (MOD1), the variables a_{ijk} and b_{ijm} were pre-fixed according to the conditions:

$$a_{ijk} = 0, \quad i, j, k \in N, \quad \text{if } r_{ij} < c_{ik}, \quad (60)$$

$$b_{ijm} = 0, \quad i, j, m \in N, \quad \text{if } r_{ij} < c_{mj}. \quad (61)$$

In the mathematical formulation (15)–(22), the variables y_{ijkm} have been pre-fixed, using the same condition applied by [Taherkhani and Alumur \(2018\)](#):

$$y_{ijkm} = 0, \quad i, j, k, m \in N, \quad \text{if } r_{ij} < c_{ik} + c_{mj}. \quad (62)$$

In the tests using the CPLEX's Benders decomposition method, the integrality of the variables y_{ijkm} , $i, j, k, m \in N$, of the formulation (MOD2) is relaxed. Thus, the Benders decomposition applied by CPLEX attains better performance.

5.1.2. Results with CAB data set

The US Civil Aeronautics Board (CAB) data set ([O'Kelly, 1987](#)) refers to airline passenger interactions between 25 US cities in 1970. This data set, which was also used by [Taherkhani and Alumur \(2018\)](#), provides transport costs (c_{ij}) and demand (w_{ij}) between each pair of cities, the demand matrix being symmetric. Each demand value was divided by $\sum_{i \in N} \sum_{j \in N} w_{ij}$, so that the total sum of the demand was equal to one.

As proposed by [Taherkhani and Alumur \(2018\)](#), for the revenue (r_{ij}) per unit of demand met between each pair of origin and destination, three values were used: 1000, 1500, and 2000. These values are referred to as low, medium, and high revenue, respectively. The revenue per unit of demand is independent of the pair of nodes served. The cost of installing the hub nodes in the experiments is $s_k \in \{50, 100, 150\}$. These values are referred to, similar to the revenue, as low, medium, and high installation costs, respectively.

The installation cost of a hub arc was taken to be 10% of the installation cost of hubs: 5 (low), 10 (medium), and 15 (high). The revenue and fixed costs for installing hubs and hub arcs were the

same used by [Taherkhani and Alumur \(2018\)](#). The discount factors considered is $\alpha \in \{0.2, 0.4, 0.6, 0.8\}$.

The results of the computational experiments carried out with both formulations are shown in Table 2. This table presents the maximum profit, the percentage of origin and destination node pairs that were served, the percentage of satisfied demand, and the hub locations. Additionally, for each formulation, this table presents the CPU time (in seconds) to solve each instance by the CPLEX with the default configuration and by the CPLEX with its Benders decomposition algorithm activated (BD-cpx), and the linear programming relaxation gap (LP Gap). The percentage of pairs served was obtained by dividing the number of pairs served by 600, which corresponds to the total number of pairs of nodes in the CAB data set and multiplying the results by 100. The percentage of satisfied demand refers to the amount of demand from the pairs of nodes in the network that has been met in relation to the total amount of demand. It was calculated by adding the fraction of the demand for each pair of nodes that was satisfied, divided by the total sum of the demand for all pairs of nodes in the network, multiplied by 100. The LP Gap was calculated by the quotient $\frac{100(l-l^*)}{l^*}$, where l^* is the objective function value of the optimal integer solution and l is the objective function value of the linear programming relaxation optimal solution. The column Costs refers to the hub installation costs and hub arc installation costs.

Table 2 shows that an increase in both fixed costs and the value of α results in lower profit, lower percentages of pairs served, and lower demand satisfied. For both formulations, the CPU time to solve the instances is lower for instances with higher fixed costs and fewer economies of scale, i.e., higher values for α . Note that the number of installed hubs decreases or remains the same for this kind of instance. When decreasing the amount of revenue, the profit and the percentages of pairs served, and the satisfied demands also decrease. It can also be noted that an increase in revenue levels, in general, increases the number of installed hubs.

Fig. 1 shows two solutions obtained by the formulations, in which the triangles represent the installed hubs, and the bold segments indicate hub arcs in which economies of scale have been achieved. Fig. 1a shows the obtained solution considering low revenue and costs, and $\alpha = 0.8$. It illustrates the case where only 31.33% of the pairs of nodes were served and shows an interesting situation. Although two hubs are installed, and there is a segment between them, no economies of scale are applied to the solution, since there is no hub arc installed on the network. This fact is due to the discount factor $\alpha = 0.8$ generates a low economy of scale and is not profitable in this case. Fig. 1b corresponds to the scenario in which revenue is high, the fixed costs are low, and $\alpha = 0.2$. It displays the solution in which the demand between all pairs of nodes has been met. Note that the hub network is incomplete.

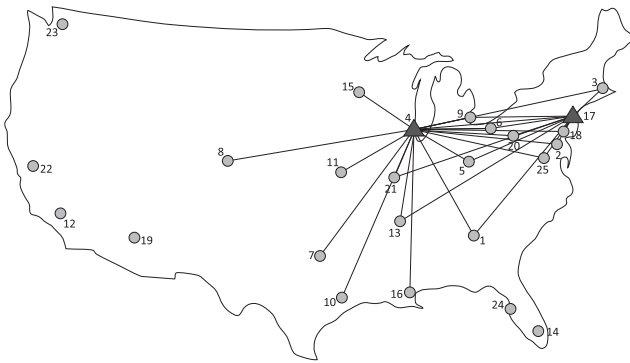
Two metrics are used to compare the two formulations: (i) CPU time and (ii) LP Gap. Table 2 shows that, for lower fixed costs, the CPU times obtained by the CPLEX and the BD-cpx using the formulation (MOD1) are less than the CPU times attained with the formulation (MOD2). On the other hand, for higher fixed costs, the opposite situation occurs (see, for example, the shortest CPU times, for each instance, highlighted in bold). Furthermore, most instances for which using the formulation (MOD2) resulted in shorter CPU times refer to optimal solutions with simple hub networks, which select to install no more than two hubs. These instances are characterized by a combination of parameters, such as low economy of scale, high fixed cost, or low revenue, which induce the installation of few hubs and hub arcs. Note that, for this set of instances, there is little difference in the CPU times attained using the two formulations. However, instances for which was obtained shorter CPU times using the formulation (MOD1) result in more complex hub networks, with more than two hubs, and the difference in the CPU times is, in general, quite expressive.

Figs. 2 and 3 show a graphical visualization of the CPU times, considering a logarithmic scale (with logarithm in base 10), due to the

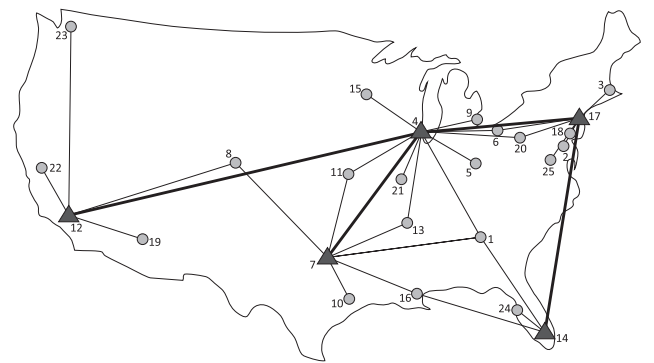
Table 2

Comparison of our formulation with the multiple allocation model of [Taherkhani and Alumur \(2018\)](#), with the CAB data set.

Costs	α	Profit	Pairs served (%)	Satisfied demand (%)	Hub locations	Formulation (MOD1)			Multiple allocation model (MOD2)		
						CPLEX time (s)	BD-cpx time (s)	LP Gap (%)	CPLEX time (s)	BD-cpx time (s)	LP gap (%)
High revenue											
Low	0.2	1162.92	100.00	100.00	4, 7, 12, 14, 17	4605.97	94.14	0.00	98,536.32	1092.15	3.29
	0.4	1008.46	97.00	99.27	4, 7, 12, 14, 17	4388.60	109.18	0.00	321,492.74	4036.69	4.82
	0.6	898.24	90.67	96.83	1, 4, 12, 17	2046.60	57.74	0.00	209,938.40	772.20	3.91
	0.8	839.40	86.67	92.38	1, 4, 12, 17	620.47	36.63	0.13	40,813.36	114.65	2.43
Medium	0.2	911.27	99.00	99.77	4, 12, 17, 24	7366.81	167.39	0.00	10,391.64	774.92	4.59
	0.4	803.73	92.67	97.66	4, 12, 17	2963.01	95.01	0.00	936.55	148.26	3.20
	0.6	717.73	88.67	95.80	4, 12, 17	1639.54	47.95	0.00	3,410.04	66.63	3.81
	0.8	690.90	79.00	78.87	18, 21	382.43	33.16	0.00	317.74	25.90	0.63
High	0.2	738.08	97.00	98.72	4, 12, 17	6855.42	265.28	0.00	640.09	99.98	4.90
	0.4	633.73	92.67	97.66	4, 12, 17	4441.65	87.30	0.00	780.24	85.28	6.05
	0.6	599.18	66.67	70.63	20	1478.37	50.01	0.00	85.40	36.47	0.55
	0.8	599.18	66.67	70.63	20	330.08	26.71	0.00	33.22	16.36	0.00
Medium revenue											
Low	0.2	665.79	94.00	98.62	4, 7, 12, 14, 17	5135.27	149.07	0.00	18,029.76	2303.90	5.69
	0.4	520.25	82.33	92.44	4, 12, 14, 17	3201.77	122.46	0.00	33,363.71	2538.85	8.83
	0.6	439.14	68.33	82.75	4, 12, 17	1513.29	48.07	0.00	28,594.78	848.08	6.40
	0.8	424.73	61.00	67.71	4, 18	330.63	27.34	0.00	4,019.81	27.75	1.59
Medium	0.2	426.89	83.00	92.28	4, 12, 17	5836.47	109.90	0.00	3,338.42	321.75	7.75
	0.4	348.40	63.00	68.83	4, 17	2993.14	65.57	0.00	167.71	34.65	3.96
	0.6	327.82	61.33	68.29	4, 17	774.59	34.18	0.00	237.21	21.74	4.50
	0.8	324.73	61.00	67.71	4, 18	271.72	27.59	0.00	57.55	9.74	1.22
High	0.2	266.41	63.00	68.83	4, 17	7922.50	139.32	0.00	892.24	50.50	9.84
	0.4	259.89	52.67	63.20	20	2069.65	32.43	0.00	40.56	16.66	0.00
	0.6	259.89	52.67	63.20	20	428.45	24.17	0.00	38.69	9.85	0.00
	0.8	259.89	52.67	63.20	20	154.05	14.11	0.00	35.43	7.35	0.00
Low revenue											
Low	0.2	197.97	58.67	80.92	4, 12, 14, 17	3932.77	152.82	0.00	3,425.26	581.67	12.66
	0.4	156.90	35.33	48.58	4, 17	832.17	32.35	0.00	32.65	33.73	4.61
	0.6	141.69	33.33	48.19	4, 17	159.85	16.04	0.00	27.49	13.88	5.40
	0.8	132.16	31.33	46.57	4, 17	61.69	11.48	0.00	24.82	6.44	4.21
Medium	0.2	69.02	36.00	49.79	4, 17	3300.05	39.01	0.00	29.26	28.51	19.34
	0.4	65.28	16.00	33.63	17	721.27	11.64	0.00	23.92	15.72	0.00
	0.6	65.28	16.00	33.63	17	160.90	8.11	0.00	20.81	5.62	0.00
	0.8	65.28	16.00	33.63	17	41.50	6.51	0.00	20.42	4.05	0.00
High	0.2	15.28	16.00	33.63	17	2915.96	23.18	0.00	10.79	8.21	0.00
	0.4	15.28	16.00	33.63	17	619.81	8.85	0.00	10.30	4.59	0.00
	0.6	15.28	16.00	33.63	17	114.51	7.64	0.00	10.42	4.14	0.00
	0.8	15.28	16.00	33.63	17	35.81	6.96	0.00	11.09	3.27	0.00



(a) Low revenue, low costs and $\alpha = 0.8$



(b) High revenue, low costs and $\alpha = 0.2$

Fig. 1. Two solutions obtained by the formulations.

difference between the orders of magnitude of the obtained values. [Fig. 2](#) shows the runtimes attained by the CPLEX and shows the situation described above, in which the runtimes spent by the proposed formulation are much shorter for the most challenging instances. For example, with high revenue, low cost, and $\alpha = 0.4$, the CPU times associated with formulation (MOD1) was 70 times shorter than the CPU times of the other model. [Fig. 3](#) shows the CPU times spent by the solver's Benders decomposition method. In this figure, for instances

with high and low costs, the CPU times attained with the formulation proposed in this article are relatively close to the CPU times obtained with the other model. Again, for the more complex instances, corresponding to low costs, the runtimes obtained with formulation (MOD1) are shorter than those presented with the multiple allocation model.

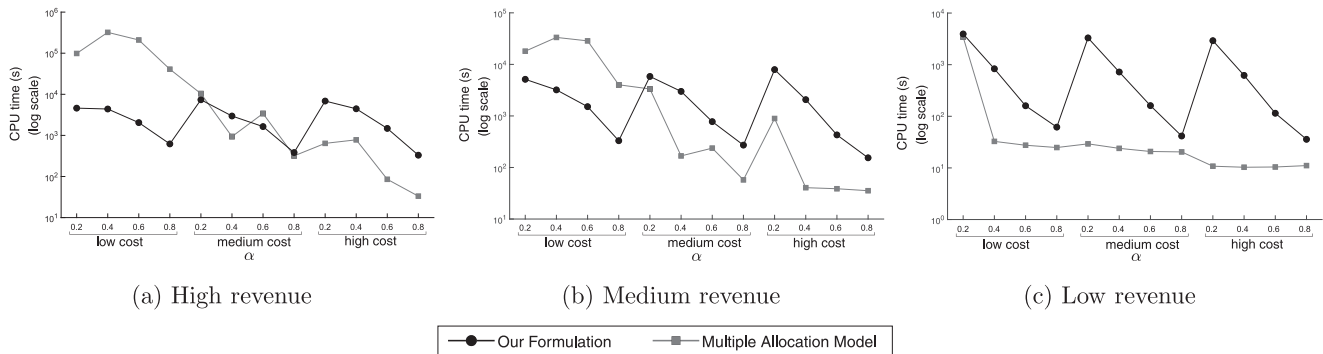
Regarding the LP gap values, the results show that formulation (MOD1) stands out by providing a null gap values for practically all instances, thus demonstrating, as expected, to be a formulation with

Table 3

Comparison of our (MOD1) formulation with the multiple allocation model of the (MOD2) formulation from [Taherkhani and Alumur \(2018\)](#), with 40-node instances of the AP data set.

Instance	Revenue	α	Profit	Pairs served (%)	Satisfied demand (%)	Hub locations	Formulation (MOD1)			Multiple allocation model (MOD2)		
							CPLEX time (s)	BD-cpx time (s)	LP gap (%)	CPLEX time (s)	BD-cpx time (s)	LP gap (%)
40L	20	0.2	17,531.46	64.19	79.98	12, 22, 26, 28	mem	6794.29	0.00	time (6.14%)	time (9.75%)	0.09
40L	20	0.4	14,635.61	48.44	67.83	11, 22, 28	mem	1268.10	0.00	time (0.05%)	2989.00	0.06
40L	20	0.6	14,099.19	22.63	45.59	28	mem	368.24	0.00	294.75	128.82	0.00
40L	20	0.8	14,099.19	22.63	45.59	28	mem	148.61	0.00	121.85	31.12	0.00
40T	20	0.2	6404.58	23.50	46.26	29	mem	601.17	0.00	232.52	174.87	0.13
40T	20	0.4	6404.58	23.50	46.26	29	mem	130.32	0.00	115.11	38.84	0.00
40T	20	0.6	6404.58	23.50	46.26	29	mem	75.86	0.00	98.44	23.87	0.00
40T	20	0.8	6404.58	23.50	46.26	29	mem	88.90	0.00	84.33	18.65	0.00
40L	30	0.2	52,299.43	84.06	91.78	12, 22, 26, 28	mem	3502.71	0.00	time (4.85%)	time (6.50%)	0.04
40L	30	0.4	47,475.40	81.63	90.27	12, 22, 26, 28	mem	1122.06	0.00	time (3.63%)	time (7.38%)	0.04
40L	30	0.6	44,213.55	76.81	87.52	11, 22, 28	mem	1112.99	0.00	time (3.19%)	10,161.22	0.04
40L	30	0.8	42,990.07	74.44	86.40	11, 22, 28	mem	329.24	0.00	time (2.58%)	159.54	0.01
40T	30	0.2	36,135.48	73.94	84.76	14, 29	mem	1884.99	0.00	time (0.00%)	903.79	0.05
40T	30	0.4	33,918.82	71.81	83.83	14, 29	mem	830.30	0.00	4320.12	242.87	0.03
40T	30	0.6	32,709.49	68.81	81.74	14, 29	mem	432.24	0.00	9768.54	94.33	0.02
40T	30	0.8	32,709.49	68.81	81.74	14, 29	mem	351.67	0.00	383.52	63.70	0.00
40L	50	0.2	130,906.36	100.00	100.00	3, 7, 11, 22, 25, 28	mem	723.75	0.00	time (2.07%)	time (4.45%)	0.02
40L	50	0.4	124,056.22	98.81	99.61	12, 22, 26, 28	mem	802.11	0.00	time (4.23%)	time (3.48%)	0.02
40L	50	0.6	120,056.17	98.38	99.52	12, 22, 26, 28	mem	472.81	0.00	time (1.80%)	time (1.28%)	0.02
40L	50	0.8	118,124.06	97.94	99.43	12, 22, 26, 28	mem	284.74	0.00	time (1.75%)	1156.56	0.01
40T	50	0.2	111,734.03	98.88	99.49	11, 22, 29	mem	965.49	0.00	time (2.99%)	8520.05	0.03
40T	50	0.4	107,847.14	96.25	98.57	14, 29	mem	757.10	0.00	time (1.43%)	1124.52	0.02
40T	50	0.6	105,857.00	95.13	98.23	14, 29	mem	357.87	0.00	time (1.34%)	459.42	0.02
40T	50	0.8	105,603.50	95.13	98.23	14, 29	mem	264.70	0.00	time (0.56%)	217.55	0.00

Notes: (i) time: time limit exceeded (14,400 s); (ii) mem: memory exceeded (16 GB).

**Fig. 2.** CPU time recorded by CPLEX for different parameter settings.

stronger linear relaxation than the other model. This characteristic is reflected in the good results obtained with the CPLEX's Benders decomposition method.

Furthermore, the CPU times spent by the BD-cpx is lower than the CPU times spent by the CPLEX, which motivates the proposal of algorithms based on the Benders decomposition to solve instances of the problem.

5.1.3. Results with AP data set

The Australian Post (AP) ([Ernst and Krishnamoorthy, 1996](#)) data set contains large instances with up to 200 nodes, which represent postal code districts. The AP data set provides a nonsymmetric demand matrix and the coordinates of the nodes, which are used to compute the euclidean distances. In this case, the transportation costs c_{ij} are equal to the euclidean distances between the pairs of nodes. The AP data set also provides two sets of values for the fixed costs of installing hubs, referred to as loose (L) and tight (T). The discount factors used are $\alpha \in \{0.2, 0.4, 0.6, 0.8\}$.

As suggested by [Taherkhani and Alumur \(2018\)](#), the AP instances, when applied to the PMMAHNDP, are complemented as follows. Three

values of revenues (r_{ij}) between each pair of origin and destination are addressed: 20, 30, and 50. The fixed costs of installing hubs considered in the experiments are equal to the fixed costs provided in the AP data set divided by 10, and the fixed costs of installation of the hub arcs were taken equal to 10% of the average of the fixed costs given in the AP data set. In the experiments in this section, instances of the AP data set containing 40 nodes were used, as performed by [Taherkhani and Alumur \(2018\)](#).

Table 3 shows the results obtained by the formulations (MOD1) and (MOD2), using the instances of the AP data set with 40 nodes. In this test, the maximum time of 4 h was considered. The first three columns show the instances, revenue, and discount factor value. Analogous to **Table 2**, the next columns show the profit, the percentage of pairs served, the percentage of satisfied demand and the location of the hubs. Finally, for each of the formulations, the CPU time (in seconds) spent by CPLEX and its Benders decomposition method is reported, followed by the linear relaxation gap.

As already observed in the experiment with the CAB instances, **Table 3** shows that decreasing the value of α causes an increase in the

Table 4
Comparison of the single-cut and multiple-cuts versions.

Instance	Revenue	BD			BD- n			BD- n^2		
		Opt	Time (s)	Iter	Opt	Time (s)	Iter	Opt	Time (s)	Iter
10L	20	4/4	283.04	133.00	3/4	5,519.03	22.00	4/4	4.40	10.00
10L	30	3/4	5,664.84	261.67	3/4	5,525.79	29.67	4/4	8.89	14.00
10L	50	3/4	5,726.79	563.00	2/4	10,822.93	23.00	4/4	106.69	16.50
10T	20	4/4	988.98	112.75	3/4	5,407.69	18.00	4/4	8.42	12.50
10T	30	3/4	5,509.43	240.00	3/4	5,509.14	33.00	4/4	14.88	14.75
10T	50	3/4	5,533.60	517.00	3/4	6,662.16	47.33	4/4	84.96	17.00
20L	20	0/4	21,600.00	–	3/4	5,531.69	20.00	4/4	85.18	9.75
20L	30	0/4	21,600.00	–	2/4	10,954.83	25.00	4/4	758.33	13.25
20L	50	0/4	21,600.00	–	2/4	11,364.51	41.00	4/4	959.92	16.00
20T	20	0/4	21,600.00	–	4/4	1,654.44	18.00	4/4	20.43	8.25
20T	30	0/4	21,600.00	–	3/4	6,325.48	41.33	4/4	557.15	14.50
20T	50	0/4	21,600.00	–	2/4	10,914.63	32.00	4/4	716.58	15.75
25L	20	0/4	21,600.00	–	3/4	7,335.49	29.33	4/4	877.60	11.25
25L	30	0/4	21,600.00	–	2/4	12,666.17	45.50	3/4	6,814.71	13.00
25L	50	0/4	21,600.00	–	1/4	16,361.56	37.00	3/4	10,489.92	16.00
25T	20	0/4	21,600.00	–	4/4	5,395.29	8.00	4/4	36.13	7.75
25T	30	0/4	21,600.00	–	3/4	6,045.06	32.67	4/4	732.44	14.25
25T	50	0/4	21,600.00	–	3/4	9,439.67	51.33	4/4	4,511.05	18.00
40L	20	0/4	21,600.00	–	2/4	10,859.67	15.50	3/4	6,164.20	9.00
40L	30	0/4	21,600.00	–	1/4	16,384.17	31.00	2/4	12,458.93	12.00
40L	50	0/4	21,600.00	–	1/4	18,335.87	48.00	1/4	16,335.46	12.00
40T	20	0/4	21,600.00	–	4/4	5,312.35	24.75	4/4	854.47	9.00
40T	30	0/4	21,600.00	–	3/4	7,008.97	24.00	3/4	6,039.98	9.67
40T	50	0/4	21,600.00	–	2/4	11,421.10	32.00	2/4	11,145.20	13.50
Average		20/96	17,187.78	304.57	62/96	8,864.90	30.39	85/96	3,324.41	12.82

Note. The symbol “–” means the time limit was exceeded.

Table 5
Inclusion of Pareto-optimal cuts in the version with n^2 cuts.

Instance	Revenue	BD- n^2			BDMW- n^2		
		Opt	Time (s)	Iter	Opt	Time (s)	Iter
10L	20	4/4	4.40	10.00	4/4	0.31	5.00
10L	30	4/4	8.89	14.00	4/4	0.18	4.25
10L	50	4/4	106.69	16.50	4/4	0.25	4.75
10T	20	4/4	8.42	12.50	4/4	0.37	6.25
10T	30	4/4	14.88	14.75	4/4	0.38	6.00
10T	50	4/4	84.96	17.00	4/4	0.41	6.25
20L	20	4/4	85.18	9.75	4/4	2.34	5.50
20L	30	4/4	758.33	13.25	4/4	4.99	6.50
20L	50	4/4	959.92	16.00	4/4	5.94	7.25
20T	20	4/4	20.43	8.25	4/4	1.74	5.75
20T	30	4/4	557.15	14.50	4/4	4.59	7.00
20T	50	4/4	716.58	15.75	4/4	10.70	7.50
25L	20	4/4	877.60	11.25	4/4	5.30	6.25
25L	30	3/4	6,814.71	13.00	4/4	9.42	7.50
25L	50	3/4	10,489.92	16.00	4/4	9.75	7.25
25T	20	4/4	36.13	7.75	4/4	2.86	4.75
25T	30	4/4	732.44	14.25	4/4	9.43	7.50
25T	50	4/4	4,511.05	18.00	4/4	15.86	7.75
40L	20	3/4	6,164.20	9.00	4/4	38.80	6.75
40L	30	2/4	12,458.93	12.00	4/4	154.83	8.75
40L	50	1/4	16,335.46	12.00	4/4	205.07	10.25
40T	20	4/4	854.47	9.00	4/4	23.00	6.00
40T	30	3/4	6,039.98	9.67	4/4	34.46	7.25
40T	50	2/4	11,145.20	13.50	4/4	69.07	9.75
Average		85/96	3,324.41	12.82	96/96	25.42	6.74

value of the objective function, in the percentages of served pairs and demands satisfied, and in the number of installed hubs in the network. Furthermore, this table also allows us to conclude that instances with tight costs generally contain a smaller number of installed hubs and do not change with the variation of the constant discount factor in the hub arcs.

Concerning the performance of the formulations, we observed that when using the CPLEX solver, the (MOD1) formulation exceeded the limit of 16 GB of memory available for all instances, and the (MOD2) formulation managed to solve only 37.5% of the cases within the

maximum time of 14,400 s. With CPLEX Benders, our (MOD1) formulation was able to resolve all instances within the time limit, while the (MOD2) formulation timed out this limit by 6 instances. The best computational times recorded with CPLEX Benders are highlighted in bold. Regarding the linear relaxation gap, we see that both formulations had low values, and the (MOD1) formulation registered a null value for all cases.

The results obtained with this test, in particular the values registered with CPLEX Benders, indicate that the (MOD1) formulation is more favorable to the application of the Benders decomposition method than the (MOD2) formulation.

5.2. Comparison of algorithms

In this section, computational experiments were carried out to compare the proposed algorithms and verify their performance. In Section 5.2.1 the settings of parameters are presented. The description of the computational experiments and an analysis of the obtained results is given in Section 5.2.2.

5.2.1. Settings of parameters

The computational experiments in this section were performed with the Australian Post (AP) data set, including instances with up to 150 nodes. Furthermore, the same settings of parameters used in Section 5.1.3 was considered.

In addition to the 10, 20, 25, 40, 50, and 100 nodes instances available in the AP data set, the experiments also use 75, 125, and 150 nodes instances. These new instances were generated according to the process used by Mokhtar et al. (2019). The fixed costs of installing the hubs were obtained randomly, respecting the order of magnitude of the costs of the instances immediately below the considered instance, i.e., the fixed costs of the instances with 75 nodes were obtained randomly between the smallest and the largest value of fixed costs for 50 nodes instances. Similarly, the fixed costs of instances with size 125 were obtained with the instances of size 100, and the fixed costs for instances with 150 nodes were calculated from the fixed costs of the 125 nodes instances.

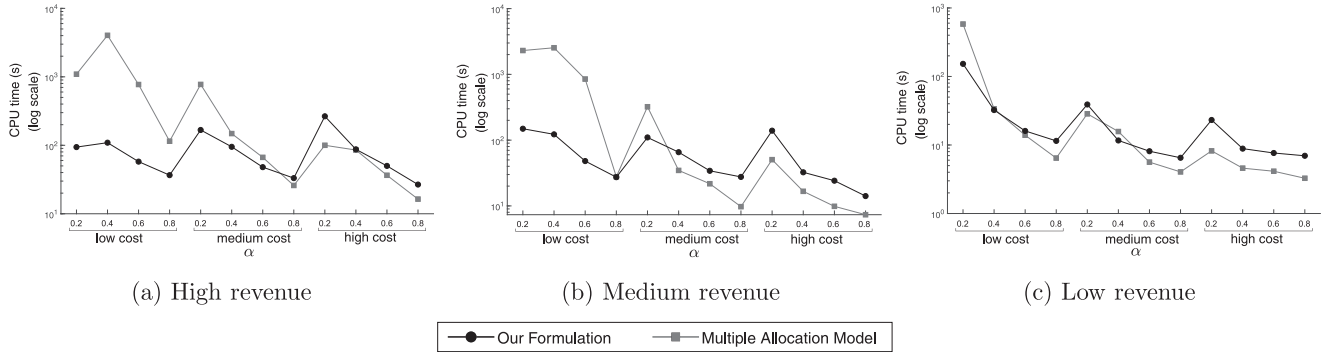


Fig. 3. CPU time recorded by Benders of CPLEX (BD-cpx) for different parameter settings.

Table 6

Comparison of four strategies for adding cuts in the branch-and-cut tree: Only for integer solutions (BDCFI), only for fractional solutions at the root node (BDCFR), for all integer solutions and fractional solutions at the root node (BDCF) and for all integer and fractional solutions (BDCFA).

Instance	Revenue	BDCFI			BDCFR			BDCF			BDCFA		
		Time (s)	Cuts	Nodes	Time (s)	Cuts	Nodes	Time (s)	Cuts	Nodes	Time (s)	Cuts	Nodes
10L	20	0.32	250.25	11.25	0.22	463.75	0.00	0.29	292.25	0.00	0.29	292.25	0.00
10L	30	0.34	278.75	7.50	0.23	498.50	0.00	0.31	307.50	0.00	0.31	307.50	0.00
10L	50	0.40	301.00	22.00	0.21	501.00	0.00	0.35	324.50	0.00	0.34	324.50	0.00
10T	20	0.46	293.00	35.00	0.32	602.75	0.00	0.49	356.75	0.25	0.48	356.75	0.25
10T	30	0.49	352.00	12.50	0.33	628.25	0.00	0.49	425.00	0.25	0.49	425.00	0.25
10T	50	0.49	363.00	4.50	0.38	636.50	0.00	0.54	447.50	1.25	0.53	452.75	0.25
20L	20	2.90	1,024.25	14.75	2.31	2,458.00	0.00	2.56	1,232.50	0.00	2.57	1,232.50	0.00
20L	30	9.98	1,655.25	292.50	5.35	3,481.50	0.75	4.51	1,959.75	0.00	4.53	1,959.75	0.00
20L	50	8.26	1,838.75	237.50	6.48	3,893.25	0.00	4.98	2,189.25	0.00	4.97	2,189.25	0.00
20T	20	2.34	951.50	0.00	1.88	2,284.25	0.00	2.35	1,023.25	0.00	2.36	1,023.25	0.00
20T	30	5.83	1,725.25	1.75	4.50	3,417.50	1.25	4.68	1,956.25	0.00	4.69	1,956.25	0.00
20T	50	9.21	2,093.25	87.25	4.72	3,357.50	4.25	4.61	2,395.25	0.00	4.62	2,395.25	0.00
25L	20	7.41	1,808.75	12.50	4.65	3,452.25	0.00	7.26	2,023.00	0.00	7.29	2,023.00	0.00
25L	30	21.53	2,693.00	90.50	8.06	5,135.00	0.00	9.66	3,021.00	0.00	9.63	3,021.00	0.00
25L	50	18.57	3,124.25	215.00	9.09	4,870.25	5.25	10.88	3,681.50	0.00	10.93	3,681.50	0.00
25T	20	3.66	1,431.25	0.00	2.70	2,765.25	0.00	3.69	1,431.25	0.00	3.68	1,431.25	0.00
25T	30	9.83	2,652.50	0.00	7.95	4,811.00	0.00	10.03	3,063.00	0.00	10.03	3,063.00	0.00
25T	50	12.07	3,200.75	7.50	8.08	5,034.00	0.00	10.19	3,664.00	0.00	10.19	3,664.00	0.00
40L	20	62.09	5,153.75	16.25	31.34	10,339.25	0.00	48.13	5,705.25	0.00	47.75	5,705.25	0.00
40L	30	146.19	8,285.75	97.50	56.94	13,636.50	0.00	96.20	9,844.75	0.00	95.64	9,844.75	0.00
40L	50	327.01	11,849.25	306.75	62.46	14,448.75	0.00	97.44	11,867.75	0.00	96.74	11,867.75	0.00
40T	20	26.14	4,456.25	0.00	22.33	9,144.75	0.00	26.33	4,521.75	0.00	26.16	4,521.75	0.00
40T	30	47.97	7,215.00	0.00	36.41	11,567.00	0.00	46.60	7,244.00	0.00	46.37	7,244.00	0.00
40T	50	75.81	9,333.25	11.50	53.76	15,481.75	0.00	62.09	10,014.75	0.00	61.60	10,014.75	0.00
Average		33.30	3,013.75	61.83	13.78	5,121.19	0.48	18.94	3,291.32	0.07	18.84	3,291.54	0.03

All experiments were performed on an Intel Core i7-7500U computer, 2.70 GHz, with 16 GB of RAM, using the Linux environment. All Benders decomposition versions were coded in C++ and the CPLEX 12.10 solver was used to solve the MP, the Magnanti-Wong independent subproblem, and the Benders subproblems of the BD, BD- n , and BD- n^2 versions, described in the next section. The MIP tolerance gap considered is equal to 10^{-7} . For the Benders decomposition algorithms, the value of M , i.e., the upper bound of η in the constraint (52), was set as the sum of the revenues, which is a valid upper bound. In fact, the value of M is not relevant to the method, since the value of this parameter does not affect the value of the binary variables in the master problem, and, after the first round of cuts is generated, it affects neither the upper bound.

5.2.2. Experiments description and analysis of results

The experiments were carried out in four stages. The results for the first two stages present the performance of several variants of the Benders decomposition method and some versions of the BBC scheme to solve the PMMAHNDP with 10 to 40 nodes instances. These first experiments were performed to find the most promising Benders decomposition version, for each approach, to solve the problem. In the third stage, the best algorithm of each approach is compared with the CPLEX solver and the CPLEX's Benders decomposition method

to evaluate the computational effort to solve instances with up to 100 nodes. In the last stage, the best algorithm pointed out in the previous experiments was used to solve instances with 125 and 150 nodes to analyze their performance when tackling large instances of the problem. A time limit of 6 h was considered for the first two stages, and a time limit of 24 h was defined for the third and fourth stages. The computational experiments for all Benders decomposition versions implemented are performed using a single thread.

The first stage aims to compare the classic version of the Benders decomposition method (BD) with two variants that use the multiple cuts strategy: BD- n , which adds n cuts in each iteration, and BD- n^2 , which adds n^2 cuts per iteration, one for each origin and destination pair. Table 4 presents a summary of the results. This table shows the number of instances solved (Opt), the average CPU time in seconds to solve the problem, and the average iteration numbers (Iter). The instances were grouped according to the four values used for the discount factor (Table B.10 of the Appendix B shows the full result).

Table 4 shows that the classic Benders method, which adds one cut by iteration, was able to solve only instances with 10 nodes, within the considered time limit. For these instances, the number of iterations spent by this version is much higher than in the versions with multiple cuts. Note that none of these approaches was able to

Table 7

Comparison of the best Benders decomposition variants and CPLEX and Benders of CPLEX using Australian Post Data Set — Part 1.

Instance	Revenue	$\alpha = 0.2$					$\alpha = 0.4$				
		CPLEX Time (s)	BD-cpx Time (s)	BD-cpx1 Time (s)	BDMW- n^2 Time (s)	BDCFR Time (s)	CPLEX Time (s)	BD-cpx Time (s)	BD-cpx1 Time (s)	BDMW- n^2 Time (s)	BDCFR Time (s)
10L	20	1.02	0.57	0.42	0.83	0.50	0.86	0.61	0.36	0.21	0.17
10L	30	1.03	0.87	0.68	0.34	0.35	0.73	0.57	0.43	0.14	0.22
10L	50	1.26	0.43	0.28	0.55	0.27	0.76	0.40	0.24	0.14	0.17
10T	20	1.85	0.85	0.73	0.85	0.53	1.18	0.63	0.49	0.23	0.23
10T	30	1.96	0.93	0.77	0.65	0.49	1.39	0.66	0.53	0.33	0.29
10T	50	2.07	0.92	0.65	0.64	0.50	1.43	0.50	0.30	0.32	0.28
20L	20	134.26	21.39	11.98	5.39	5.18	101.70	12.89	6.88	1.83	1.91
20L	30	121.90	30.03	13.68	12.09	13.02	140.94	23.00	14.68	4.84	4.39
20L	50	131.81	14.66	8.82	11.90	16.91	129.23	10.17	6.10	6.93	3.55
20T	20	133.94	12.20	6.38	3.16	3.05	72.83	7.23	4.08	1.90	1.93
20T	30	124.91	59.61	26.51	9.53	9.64	126.19	20.35	12.13	4.68	3.43
20T	50	146.60	18.34	9.91	31.79	5.90	125.18	12.96	7.32	6.66	4.51
25L	20	3,164.13	102.55	62.16	10.39	7.15	1,572.35	69.64	25.91	6.60	6.85
25L	30	4,500.43	180.94	51.35	20.16	14.88	1,888.68	73.53	30.59	7.70	6.63
25L	50	4,577.06	50.51	21.93	15.50	19.17	2,800.55	59.56	23.40	13.20	7.99
25T	20	3,284.25	25.32	9.55	4.37	3.92	733.74	17.99	8.14	2.93	2.81
25T	30	8,060.80	185.64	73.65	22.01	14.68	3,253.37	120.88	33.29	5.39	5.64
25T	50	10,722.74	89.28	37.94	42.46	17.30	4,085.72	58.22	22.49	11.39	7.08
40L	20	mem	6,794.29	1,992.74	94.70	58.35	mem	1,268.10	270.34	28.19	34.67
40L	30	mem	3,502.71	1,842.67	513.07	106.70	mem	1,122.06	672.91	51.96	53.66
40L	50	mem	723.75	412.50	451.59	74.92	mem	802.11	260.12	247.78	97.30
40T	20	mem	601.17	127.48	37.57	35.10	mem	130.32	58.84	22.67	22.51
40T	30	mem	1,884.99	823.48	53.06	55.02	mem	830.30	287.52	30.91	39.19
40T	50	mem	965.49	360.54	100.39	77.54	mem	757.10	246.94	105.85	55.77
50L	20	mem	mem	12,339.07	161.11	148.99	mem	mem	931.93	81.70	87.25
50L	30	mem	mem	5,451.71	298.94	334.89	mem	mem	2,921.75	269.33	179.65
50L	50	mem	mem	2,560.89	921.96	329.77	mem	mem	976.23	596.23	265.10
50T	20	mem	mem	429.29	64.00	64.35	mem	mem	198.49	56.42	56.35
50T	30	mem	mem	2,701.35	207.24	219.72	mem	mem	795.69	107.99	114.27
50T	50	mem	mem	1,003.68	553.88	289.18	mem	mem	735.18	206.27	165.83
75L	20	mem	mem	mem	1,605.41	2,277.41	mem	mem	mem	1,087.00	1,348.26
75L	30	mem	mem	mem	2,409.20	3,516.02	mem	mem	mem	1,973.44	1,988.41
75L	50	mem	mem	mem	8,502.26	3,464.41	mem	mem	mem	5,367.35	2,366.14
75T	20	mem	mem	mem	965.96	931.39	mem	mem	mem	629.95	627.04
75T	30	mem	mem	mem	2,142.82	2,298.20	mem	mem	mem	1,214.21	1,175.24
75T	50	mem	mem	mem	2,056.07	2,374.03	mem	mem	mem	1,545.68	1,561.35
100L	20	mem	mem	mem	8,885.35	9,427.82	mem	mem	mem	7,596.66	6,299.13
100L	30	mem	mem	mem	time (0.01%)	25,615.63	mem	mem	mem	12,684.55	6,411.47
100L	50	mem	mem	mem	time (2.41%)	21,249.75	mem	mem	mem	44,446.88	12,380.15
100T	20	mem	mem	mem	2,972.80	3,470.80	mem	mem	mem	2,459.90	2,435.28
100T	30	mem	mem	mem	5,300.22	5,981.44	mem	mem	mem	3,424.76	3,435.93
100T	50	mem	mem	mem	9,333.83	8,729.69	mem	mem	mem	3,834.67	4,295.65

Notes. Time, 24 h time limit exceeded; mem, 16 GB memory exceeded.

solve all 96 instances within the defined time limit. However, the BD- n^2 method, which adds n^2 cuts per iteration, was the version that solved the greatest number of instances. Furthermore, this version presents the shortest average runtimes and the smallest average number of iterations. For this reason, the strategy of adding n^2 cuts was adopted for all subsequent tests.

The version of the Benders decomposition that uses the Pareto-optimal cuts, shown in the Algorithm 2, is named BDMW- n^2 . The results achieved by this version are displayed in Table 5 (the complete results are available in Table B.11 of the Appendix B). This table shows that the addition of Pareto-optimal cuts significantly improves the performance of the algorithm. All instances considered in this stage were solved to optimality, and there was a significant decrease in computational time and number of iterations. This fact reinforces how much the quality of the cuts influences the performance of the Benders decomposition method.

The second set of tests was performed to compare some methods using the BBC scheme, presented in Section 4.2.3. In these methods, in general, a large number of Benders cuts are added to the nodes in the branch-and-cut tree ($O(n^2)$ cuts). To reduce the added cuts, a cut filtering scheme is applied. Instead of adding all the Benders cuts obtained from the SP dual solution, the algorithm retains only those violated by the current MP solution. Martins de Sá et al. (2015) used this filtering scheme in the computational experiments and reported a

decrease in the number of cuts added and a reduction in the CPU time spent to solve the problems. These characteristics were also observed in our experiments.

Using the cut filtering mechanism, four strategies for adding Benders cuts to the Branch-and-Bound (B&B) tree are compared. The first strategy is to add Benders cuts obtained exclusively from integer solutions (BDCFI). The second strategy adds Benders cuts only for fractional solutions at the root node (BDCFR). The combination of these two strategies constitutes the third strategy adopted (BDCF). The fourth and last strategy is to add Benders cuts at any level of the B&B tree, from any solution, whether integer or fractional (BDCFA). Table 6 shows a summary of the obtained results, in which the instances are grouped according to the values of α (detailed results are available in Table B.12 in the Appendix B). This table shows the average CPU time, in seconds, spent by the methods to solve the instances, the average number of cuts added in the branch-and-cut tree ("Cuts"), and the average number of nodes explored in the B&B tree ("Nodes").

Table 6 shows that the strategy that adds Benders cuts only from integer solutions (BDCFI) adds the least number of cuts. On the other hand, this version is the one that most exploits the nodes of the tree and demands the most computational time to solve the instances. For the other versions, the branch-and-cut procedure applied by the CPLEX remains in the root node for practically all instances, presenting an average number of exploited nodes less than one. Furthermore, the

Table 8

Comparison of the best Benders decomposition variants and CPLEX and Benders of CPLEX using Australian post data set — Part 2.

Instance	Revenue	$\alpha = 0.6$					$\alpha = 0.8$				
		CPLEX Time (s)	BD-cpx Time (s)	BD-cpx1 Time (s)	BDMW- n^2 Time (s)	BDCFR Time (s)	CPLEX Time (s)	BD-cpx Time (s)	BD-cpx1 Time (s)	BDMW- n^2 Time (s)	BDCFR Time (s)
10L	20	0.47	0.39	0.24	0.10	0.08	0.30	0.31	0.19	0.11	0.11
10L	30	0.70	0.58	0.32	0.10	0.09	0.52	0.53	0.31	0.15	0.24
10L	50	0.74	0.42	0.19	0.14	0.18	0.63	0.38	0.23	0.17	0.21
10T	20	0.79	0.60	0.46	0.24	0.31	0.62	0.49	0.35	0.16	0.20
10T	30	1.83	0.94	0.71	0.36	0.36	0.57	0.45	0.31	0.17	0.18
10T	50	1.77	1.02	0.65	0.43	0.60	0.81	0.39	0.25	0.23	0.14
20L	20	29.22	6.70	3.67	1.22	1.23	12.82	3.93	2.58	0.93	0.92
20L	30	82.55	13.24	6.78	1.76	2.22	30.39	7.42	4.22	1.27	1.77
20L	50	93.91	8.19	5.32	2.24	2.80	62.53	6.47	4.33	2.69	2.66
20T	20	22.54	4.59	2.75	0.98	1.20	13.28	4.21	2.68	0.91	1.34
20T	30	93.88	13.78	6.96	2.76	2.87	35.54	10.05	6.00	1.40	2.04
20T	50	105.60	7.25	4.82	2.84	6.91	48.80	6.91	4.98	1.50	1.57
25L	20	277.72	32.88	12.42	2.72	3.12	42.25	12.70	6.72	1.49	1.47
25L	30	870.13	48.07	34.87	5.68	6.26	357.97	38.16	14.76	4.13	4.46
25L	50	1072.55	32.96	15.81	5.92	5.15	393.82	24.02	12.85	4.36	4.03
25T	20	122.67	8.07	5.10	2.12	2.09	32.96	7.76	4.88	2.00	1.96
25T	30	993.60	51.00	22.98	5.86	6.66	346.46	42.52	20.94	4.46	4.83
25T	50	1159.56	44.55	17.01	5.93	4.49	403.03	25.99	15.33	3.66	3.43
40L	20	mem	368.24	134.12	20.11	20.11	mem	148.61	71.74	12.20	12.22
40L	30	mem	1112.99	313.34	33.08	40.06	mem	329.24	121.08	21.22	27.32
40L	50	mem	472.81	160.03	83.97	49.27	mem	284.74	103.16	36.94	28.33
40T	20	mem	75.86	35.21	16.54	16.54	mem	88.90	33.69	15.20	15.17
40T	30	mem	432.24	166.02	32.77	30.78	mem	351.67	128.58	21.09	20.66
40T	50	mem	357.87	145.76	43.77	49.38	mem	264.70	122.51	26.25	32.36
50L	20	mem	mem	452.19	50.03	50.28	mem	mem	237.06	29.86	29.77
50L	30	mem	mem	2082.21	205.96	121.95	mem	mem	572.51	94.95	227.77
50L	50	mem	mem	847.42	499.04	445.13	mem	mem	348.08	77.77	82.92
50T	20	mem	mem	104.81	40.78	40.85	mem	mem	105.09	37.27	37.27
50T	30	mem	mem	397.40	72.45	86.61	mem	mem	495.01	51.49	58.82
50T	50	mem	mem	452.05	111.50	115.27	mem	mem	382.61	64.33	61.67
75L	20	mem	mem	mem	728.64	732.49	mem	mem	mem	490.42	577.37
75L	30	mem	mem	mem	1,199.67	1215.75	mem	mem	mem	844.79	932.62
75L	50	mem	mem	mem	2,117.18	2571.40	mem	mem	mem	945.28	1108.66
75T	20	mem	mem	mem	541.62	545.76	mem	mem	mem	492.38	490.22
75T	30	mem	mem	mem	764.62	872.35	mem	mem	mem	669.03	644.32
75T	50	mem	mem	mem	1,009.15	1197.38	mem	mem	mem	722.94	818.08
100L	20	mem	mem	mem	2,453.42	2979.99	mem	mem	mem	1612.14	1874.36
100L	30	mem	mem	mem	7,873.04	4758.21	mem	mem	mem	3100.93	3290.31
100L	50	mem	mem	mem	13,704.64	6389.96	mem	mem	mem	4,487.94	3889.53
100T	20	mem	mem	mem	1,762.81	1781.70	mem	mem	mem	1,613.23	1610.26
100T	30	mem	mem	mem	2,444.66	2457.55	mem	mem	mem	2,073.76	2059.14
100T	50	mem	mem	mem	3,185.67	3667.19	mem	mem	mem	2290.08	2299.77

Note. mem, 16 GB memory exceeded.

amount of cuts added by the BDCF and BDCFA versions is the same, except for a single set of instances. Although it adds a high amount of cuts, the BDCFR algorithm, which adds cuts only from fractional solutions at the root node, presented the best average computational time. In addition, the BDCFR algorithm showed better average computational time than the BDMW- n^2 algorithm (Table 5), confirming that the inclusion of Pareto-optimal cuts within the branch-and-bound tree gives better results than the traditional version that adds the cuts only after solving the relaxed master problem.

The third stage of experiments compares the best algorithms obtained previously with the default configuration of the CPLEX solver and the CPLEX solver with its Benders decomposition algorithm activated. To better compare with our methods that make use of a single thread, a second version of the Benders algorithm of the CPLEX solver, in which the number of threads was limited to one, was also included in this stage. The results with the most challenging instances, corresponding to $\alpha = 0.2$ and $\alpha = 0.4$, are presented in Table 7, while the results with $\alpha = 0.6$ and $\alpha = 0.8$ is given in Table 8. These tables report the runtime required to solve AP instances with up to 100 nodes by CPLEX, by the CPLEX Benders decomposition method with the default configuration (BD-cpx), and by the CPLEX Benders decomposition method restricted to a single thread (BD-cpx1), and also by the two most promising methods identified in the tests previously performed, i.e., the Benders decomposition approach with

the addition of n^2 Pareto-optimal cuts per iteration (BDMW- n^2), and the BBC approach that adds Pareto-optimal cuts from fractional solutions at the root node (BDCFR). The shortest runtimes for each instance are highlighted in bold. For instances that have not been solved at optimality, the value of the gap is shown. Furthermore, “mem” indicates in the table the instances that have not been solved due to exceeding the available memory.

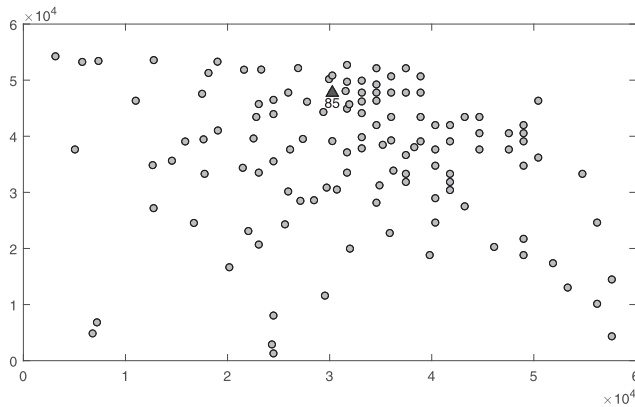
The results presented in Tables 7 and 8 show that the CPLEX solved only instances with up to 25 nodes. Using the default configuration, Benders decomposition of the CPLEX solved instances with up to 40 nodes, and, when limited to one thread, it was able to solve instances with up to 50 nodes. For all instances not solved by the solver, it exceeds the memory limit. However, concerning the instances for which the CPLEX, the BD-cpx, and the BD-cpx1 find the optimal solution, the runtime is greater than the runtime spent by the algorithms proposed in this study. On the other hand, the BDMW- n^2 method reported an effective performance, it was not able to solve only three instances, considering the time limit of 24 h. The BDCFR method obtained the best performance, solving all instances at reasonable times, and the attained longest time was approximately 12 h, which is half of the time limit.

Observe that, in all experiments performed, as the number of nodes increases, instances with fixed costs for installing hubs of type Loose require more time to be solved than instances with fixed costs for hubs of type Tight. In the literature, in which most HLPs are addressed

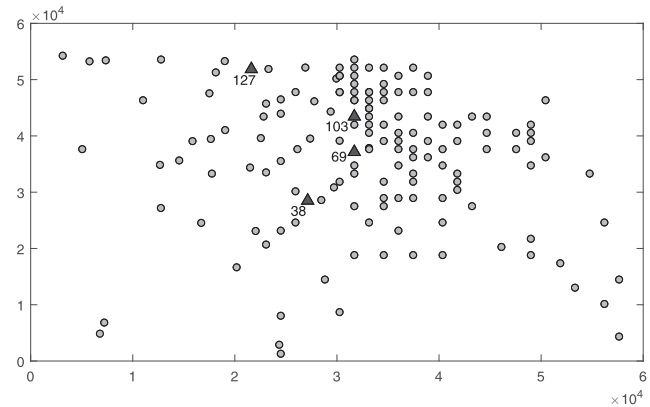
Table 9
Solutions obtained with the BDCFR method.

(a) AP instances with 125 nodes						(b) AP instances with 150 nodes					
Instance	Revenue	α	Profit	Hub locations	BDCFR Time (s)	Instance	Revenue	α	Profit	Hub locations	BDCFR Time (s)
125L	20	0.2	19,751.55	34, 66, 79, 86	26,482.87	150L	20	0.2	–	–	mem (0.01%)
125L	20	0.4	16,272.93	35, 79, 86	12,272.01	150L	20	0.4	15,557.64	38, 81, 101	30,078.62
125L	20	0.6	15,028.67	35, 79, 86	8,180.95	150L	20	0.6	14,446.68	39, 101	29,523.28
125L	20	0.8	14,966.38	35, 79, 86	5,693.27	150L	20	0.8	14,434.49	39, 101	13,859.46
125L	30	0.2	54,503.08	6, 35, 66, 79, 86	53,161.89	150L	30	0.2	–	–	mem (34.35%)
125L	30	0.4	49,301.78	34, 66, 79, 86	21,207.46	150L	30	0.4	–	–	mem (17.15%)
125L	30	0.6	45,861.48	34, 66, 79, 86	14,901.48	150L	30	0.6	45,100.57	38, 81, 93, 101	43,851.14
125L	30	0.8	44,820.26	34, 63, 79, 86	7,325.15	150L	30	0.8	43,931.25	38, 81, 94, 101	26,252.31
125L	50	0.2	–	–	mem (15.94%)	150L	50	0.2	–	–	mem (15.92%)
125L	50	0.4	126,112.19	6, 35, 66, 79, 86	28,117.28	150L	50	0.4	–	–	mem (10.89%)
125L	50	0.6	120,985.83	34, 66, 80, 86	17,584.78	150L	50	0.6	–	–	mem (0.01%)
125L	50	0.8	119,696.69	34, 63, 79, 86	10,761.85	150L	50	0.8	118,924.32	38, 81, 94, 101	29,908.91
125T	20	0.2	7,169.77	85	8,600.38	150T	20	0.2	10,913.35	103, 127	33,412.04
125T	20	0.4	7,169.77	85	6,907.87	150T	20	0.4	10,913.35	103, 127	21,305.05
125T	20	0.6	7,169.77	85	4,863.38	150T	20	0.6	10,913.35	103, 127	16,679.75
125T	20	0.8	7,169.77	85	4,306.39	150T	20	0.8	10,913.35	103, 127	15,581.67
125T	30	0.2	29,586.11	85	10,352.58	150T	30	0.2	38,270.84	69, 103, 127	64,565.21
125T	30	0.4	29,586.11	85	8,333.74	150T	30	0.4	38,270.84	69, 103, 127	25,887.18
125T	30	0.6	29,586.11	85	6,955.91	150T	30	0.6	38,270.84	69, 103, 127	17,370.94
125T	30	0.8	29,586.11	85	5,645.32	150T	30	0.8	38,270.84	69, 103, 127	14,019.36
125T	50	0.2	93,251.38	85	16,541.83	150T	50	0.2	–	–	mem (1.02%)
125T	50	0.4	93,251.38	85	11,164.57	150T	50	0.4	111,248.30	38, 69, 103, 127	35,371.05
125T	50	0.6	93,251.38	85	7,742.55	150T	50	0.6	111,248.30	38, 69, 103, 127	25,300.32
125T	50	0.8	93,251.38	85	6,421.03	150T	50	0.8	111,248.30	38, 69, 103, 127	17,870.28

Note. mem, 16 GB memory exceeded.



(a) 125 nodes



(b) 150 nodes

Fig. 4. Arrangement of the nodes of AP instances with size 125 and 150.

from a cost minimization point of view, the opposite occurs. The most challenging instances of the AP data set, in general, are those associated with Tight costs (Mokhtar et al., 2019). This particularity reported with the AP instances, when addressing the problems with profit maximization, is also observed in the results presented by Taherkhani and Alumur (2018), in which AP instances with up to 40 nodes and $\alpha = 0.75$ were used.

Finally, the tests of the last stage evaluated the performance of the BDCFR method for solving instances with 125 and 150 nodes, considering the computational limitations. Table 9 shows the obtained results and presents the gap for instances that have not been solved within the time limit. For the solved instances, the maximum profit value and the location of the hubs are also presented. According to this table, our method solved almost all problems with up to 125 nodes, except

for a single instance corresponding to $\alpha = 0.2$, Loose fixed installation costs, and revenue equal to 50. Additionally, this method solved most instances with 150 nodes, especially those with fixed costs Tight.

Table 9 shows that optimal solutions for AP instances have similar behavior to optimal solutions for CAB instances when varying economies of scale and revenue value. The increase in the revenue or in the discount in the hub arcs (i.e., a lower α value) provides the solution with a greater number of hubs in the network. In Tight cost solutions where two or more hubs were installed in the network, we observed that no hub arc was installed. This happens because the cost for installing the hub arcs is not offset by their use.

Fig. 4 shows the arrangement of the nodes of the 125 and 150 size instances, where the locations of the hubs obtained in the Tight cost solutions were highlighted. Notice in Fig. 4a that the node 85, which is

the only hub installed on all 125T instances, is located in a region with high node density. Also, the amount of demand coming in and going out of that node represents almost 30% of the total network demand. According to Ernst and Krishnamoorthy (1996), this point represents a central business district. This justifies the installation of this single hub, regardless of the value of α and revenue, since, for these instances, the costs are tighter and do not favor the installation of hubs and hub arcs in the network. On the other hand, the node with the highest inflow and outflow of demand is not always installed. The 150T instances illustrate this situation, where the node that receives and sends the greatest amount of demand has not been installed. See Fig. 4b that the installed hubs cover a good part of the region that contains the largest number of nodes. In this case, from a profit point of view, this configuration was more advantageous compared to installing a single hub, as was the case of 125T instances.

6. Conclusions

This work addressed the profit maximization HLP with a focus on maximizing profit and in designing an incomplete network, which consists of locating hubs, locating the hub arcs, and allocating the demand nodes to the hubs to maximize the total profit. Unlike the classic problems in the area of hub location, this problem has the particularity that meeting all demands is not mandatory, being carried out only for the origin and destination pairs that generate profit.

A new alternative formulation for the problem was presented and it is proved to be stronger than another model in the literature. A comparative analysis of the two formulations using 25 nodes CAB instances was also carried out. The results showed that the proposed formulation has a better performance in the face of the most difficult instances, which correspond to the problems with a greater discount applied in the hub arcs and, consequently, more favorable to the installation of hubs in the network, while the other formulation has a better performance for most instances where the optimal solution has only one or two hubs. In addition, the proposed model has better linear relaxation, presenting a null gap value for almost all CAB instances tested.

Algorithms based on the Benders decomposition method have been proposed to solve the problem. Several strategies to improve the performance of the classic version of Benders were addressed. In particular, two variants were shown to be more effective: (i) Benders algorithm with multiple cuts and addition of Pareto-optimal cuts from Magnanti–Wong points; and (ii) application of a Benders Branch-and-Cut scheme, in which Pareto-optimal cuts, also obtained through Magnanti–Wong points, are added to the Branch-and-Cut tree from fractional solutions at the root node.

Several computational experiments were realized to verify the performance of the proposed methods, considering different configurations for the problem parameters. The algorithms were also compared with the CPLEX solver and the CPLEX's Benders decomposition method. The results showed that our best versions were able to solve within the time limit the double of instances solved by the CPLEX's Benders approach and four times more instances than the CPLEX with disabling Benders decomposition. Furthermore, the best version identified, which incorporates the decomposition of Benders in a Branch-and-Cut framework, was able to solve at optimality even the AP instances with up to 150 nodes. These instances are the largest portrayed in the literature so far for the problem of hub location with maximizing profit, in which no assumption is imposed on the hub-level network.

Table B.10

Comparison of single-cut and multiple-cuts versions.

Instance	Revenue	α	BD		BD-n		BD-n ²	
			Time (s)	Iter	Time (s)	Iter	Time (s)	Iter
10L	20	0.2	1,118.80	388	time	time	14.93	19
10L	20	0.4	10.09	84	475.00	46	2.16	10
10L	20	0.6	2.35	40	0.81	12	0.34	6
10L	20	0.8	0.90	20	0.31	8	0.16	5
10L	30	0.2	time	time	time	time	29.31	22
10L	30	0.4	1,030.00	555	480.48	46	3.57	13
10L	30	0.6	21.65	143	21.24	27	1.90	12
10L	30	0.8	7.71	87	1.44	16	0.76	9
10L	50	0.2	time	time	time	time	417.90	29
10L	50	0.4	950.48	712	time	time	5.56	15
10L	50	0.6	71.40	317	90.52	31	2.41	12
10L	50	0.8	285.26	660	1.20	15	0.89	10
10T	20	0.2	3,943.27	305	time	time	29.45	21
10T	20	0.4	8.08	73	28.92	27	3.20	13
10T	20	0.6	2.37	41	1.41	17	0.70	9
10T	20	0.8	2.20	32	0.43	10	0.32	7
10T	30	0.2	time	time	time	time	47.70	22
10T	30	0.4	401.76	443	417.60	51	8.81	17
10T	30	0.6	28.50	189	18.15	36	2.62	13
10T	30	0.8	7.45	88	0.81	12	0.40	7
10T	50	0.2	time	time	time	time	322.82	28
10T	50	0.4	228.56	516	4,966.38	77	12.41	17
10T	50	0.6	134.29	479	80.90	48	4.05	15
10T	50	0.8	171.55	556	1.37	17	0.57	8
20L	20	0.2	time	time	time	time	293.51	14
20L	20	0.4	time	time	514.25	36	39.43	10
20L	20	0.6	time	time	10.30	16	5.85	8
20L	20	0.8	time	time	2.20	8	1.93	7
20L	30	0.2	time	time	time	time	2,385.08	19
20L	30	0.4	time	time	time	time	583.60	15
20L	30	0.6	time	time	608.81	37	58.99	11
20L	30	0.8	time	time	10.52	13	5.65	8
20L	50	0.2	time	time	time	time	2,632.35	21
20L	50	0.4	time	time	time	time	1,041.71	18
20L	50	0.6	time	time	2,198.37	55	145.72	14
20L	50	0.8	time	time	59.68	27	19.90	11
20T	20	0.2	time	time	6,598.23	36	67.71	11
20T	20	0.4	time	time	14.07	17	9.22	8
20T	20	0.6	time	time	3.06	10	3.11	7
20T	20	0.8	time	time	2.39	9	1.67	7
20T	30	0.2	time	time	time	time	1,917.79	20
20T	30	0.4	time	time	3,577.31	66	272.62	15
20T	30	0.6	time	time	112.19	37	30.89	13
20T	30	0.8	time	time	12.43	21	7.31	10
20T	50	0.2	time	time	time	time	2,244.52	25
20T	50	0.4	time	time	time	time	563.47	16
20T	50	0.6	time	time	439.86	43	51.38	13
20T	50	0.8	time	time	18.65	21	6.95	9
25L	20	0.2	time	time	time	time	3,174.05	14
25L	20	0.4	time	time	7,713.95	62	316.82	15
25L	20	0.6	time	time	22.97	17	15.37	9
25L	20	0.8	time	time	5.05	9	4.16	7
25L	30	0.2	time	time	time	time	time	time
25L	30	0.4	time	time	time	time	5,419.71	16
25L	30	0.6	time	time	7,349.30	60	210.08	13
25L	30	0.8	time	time	115.37	31	29.03	10
25L	50	0.2	time	time	time	time	time	time
25L	50	0.4	time	time	time	time	19,480.50	20
25L	50	0.6	time	time	time	time	802.04	15
25L	50	0.8	time	time	646.24	37	77.12	13
25T	20	0.2	time	time	21,561.84	2	122.06	11
25T	20	0.4	time	time	9.93	12	15.13	8
25T	20	0.6	time	time	5.21	9	4.30	6
25T	20	0.8	time	time	4.19	9	3.04	6
25T	30	0.2	time	time	time	time	2,468.10	19
25T	30	0.4	time	time	2,467.79	48	351.44	14
25T	30	0.6	time	time	93.56	31	84.74	13
25T	30	0.8	time	time	18.89	19	25.47	11
25T	50	0.2	time	time	time	time	16,754.70	27
25T	50	0.4	time	time	15,340.84	86	1,113.75	19
25T	50	0.6	time	time	776.62	43	146.52	14

(continued on next page)

Table B.10 (continued).

Instance	Revenue	α	BD		BD- n		BD- n^2	
			Time (s)	Iter	Time (s)	Iter	Time (s)	Iter
25T	50	0.8	time	time	41.22	25	29.22	12
40L	20	0.2	time	time	time	time	time	time
40L	20	0.4	time	time	time	time	2,830.11	12
40L	20	0.6	time	time	201.17	20	201.27	9
40L	20	0.8	time	time	37.49	11	25.41	6
40L	30	0.2	time	time	time	time	time	time
40L	30	0.4	time	time	time	time	time	time
40L	30	0.6	time	time	time	time	6,378.37	14
40L	30	0.8	time	time	736.67	31	257.35	10
40L	50	0.2	time	time	time	time	time	time
40L	50	0.4	time	time	time	time	time	time
40L	50	0.6	time	time	time	time	time	time
40L	50	0.8	time	time	8,543.49	48	541.82	12
40T	20	0.2	time	time	21,146.45	72	3,209.89	16
40T	20	0.4	time	time	62.03	13	155.79	9
40T	20	0.6	time	time	21.33	7	36.39	6
40T	20	0.8	time	time	19.57	7	15.81	5
40T	30	0.2	time	time	time	time	time	time
40T	30	0.4	time	time	6,269.42	41	2,359.83	13
40T	30	0.6	time	time	127.42	20	152.42	9
40T	30	0.8	time	time	39.03	11	47.68	7
40T	50	0.2	time	time	time	time	time	time
40T	50	0.4	time	time	time	time	time	time
40T	50	0.6	time	time	2,384.76	44	1,255.49	16
40T	50	0.8	time	time	99.64	20	125.30	11

time: 6 h time limit exceeded.

Table B.11

Inclusion of Pareto-optimal cuts in the version with n^2 cuts.

Instance	Revenue	α	BD- n^2		BDMW- n^2	
			Time (s)	Iter	Time (s)	Iter
10L	20	0.2	14.93	19	0.83	8
10L	20	0.4	2.16	10	0.21	5
10L	20	0.6	0.34	6	0.10	3
10L	20	0.8	0.16	5	0.11	4
10L	30	0.2	29.31	22	0.34	5
10L	30	0.4	3.57	13	0.14	4
10L	30	0.6	1.90	12	0.10	3
10L	30	0.8	0.76	9	0.15	5
10L	50	0.2	417.90	29	0.55	6
10L	50	0.4	5.56	15	0.14	4
10L	50	0.6	2.41	12	0.14	4
10L	50	0.8	0.89	10	0.17	5
10T	20	0.2	29.45	21	0.85	8
10T	20	0.4	3.20	13	0.23	5
10T	20	0.6	0.70	9	0.24	7
10T	20	0.8	0.32	7	0.16	5
10T	30	0.2	47.70	22	0.65	7
10T	30	0.4	8.81	17	0.33	6
10T	30	0.6	2.62	13	0.36	6
10T	30	0.8	0.40	7	0.17	5
10T	50	0.2	322.82	28	0.64	7
10T	50	0.4	12.41	17	0.32	6
10T	50	0.6	4.05	15	0.43	7
10T	50	0.8	0.57	8	0.23	5
20L	20	0.2	293.51	14	5.39	7
20L	20	0.4	39.43	10	1.83	6
20L	20	0.6	5.85	8	1.22	5
20L	20	0.8	1.93	7	0.93	4
20L	30	0.2	2,385.08	19	12.09	8
20L	30	0.4	583.60	15	4.84	7
20L	30	0.6	58.99	11	1.76	6
20L	30	0.8	5.65	8	1.27	5
20L	50	0.2	2,632.35	21	11.90	8
20L	50	0.4	1,041.71	18	6.93	8
20L	50	0.6	145.72	14	2.24	6
20L	50	0.8	19.90	11	2.69	7

The results obtained in the computational tests show that solving large instances of the problem is not an easy task since more

Table B.11 (continued).

Instance	Revenue	α	BD- n^2		BDMW- n^2	
			Time (s)	Iter	Time (s)	Iter
20T	20	0.2	67.71	11	3.16	8
20T	20	0.4	9.22	8	1.90	7
20T	20	0.6	3.11	7	0.98	4
20T	20	0.8	1.67	7	0.91	4
20T	30	0.2	1,917.79	20	9.53	7
20T	30	0.4	272.62	15	4.68	8
20T	30	0.6	30.89	13	2.76	8
20T	30	0.8	7.31	10	1.40	5
20T	50	0.2	2,244.52	25	31.79	10
20T	50	0.4	563.47	16	6.66	8
20T	50	0.6	51.38	13	2.84	7
20T	50	0.8	6.95	9	1.50	5
25L	20	0.2	3,174.05	14	10.39	8
25L	20	0.4	316.82	15	6.60	9
25L	20	0.6	15.37	9	2.72	5
25L	20	0.8	4.16	7	1.49	3
25L	30	0.2	time	time	20.16	8
25L	30	0.4	5,419.71	16	7.70	8
25L	30	0.6	210.08	13	5.68	7
25L	30	0.8	29.03	10	4.13	7
25L	50	0.2	time	time	15.50	7
25L	50	0.4	19,480.50	20	13.20	8
25L	50	0.6	802.04	15	5.92	7
25L	50	0.8	77.12	13	4.36	7
25T	20	0.2	122.06	11	4.37	6
25T	20	0.4	15.13	8	2.93	5
25T	20	0.6	4.30	6	2.12	4
25T	20	0.8	3.04	6	2.00	4
25T	30	0.2	2,468.10	19	22.01	9
25T	30	0.4	351.44	14	5.39	6
25T	30	0.6	84.74	13	5.86	8
25T	30	0.8	25.47	11	4.46	7
25T	50	0.2	16,754.70	27	42.46	10
25T	50	0.4	1,113.75	19	11.39	8
25T	50	0.6	146.52	14	5.93	7
25T	50	0.8	29.22	12	3.66	6
40L	20	0.2	time	time	94.70	10
40L	20	0.4	2,830.11	12	28.19	7
40L	20	0.6	201.27	9	20.11	6
40L	20	0.8	25.41	6	12.20	4
40L	30	0.2	time	time	513.07	12
40L	30	0.4	time	time	51.96	9
40L	30	0.6	6,378.37	14	33.08	8
40L	30	0.8	257.35	10	21.22	6
40L	50	0.2	time	time	451.59	10
40L	50	0.4	time	time	247.78	11
40L	50	0.6	time	time	83.97	11
40L	50	0.8	541.82	12	36.94	9
40T	20	0.2	3,209.89	16	37.57	8
40T	20	0.4	155.79	9	22.67	6
40T	20	0.6	36.39	6	16.54	5
40T	20	0.8	15.81	5	15.20	5
40T	30	0.2	time	time	53.06	8
40T	30	0.4	2,359.83	13	30.91	7
40T	30	0.6	152.42	9	32.77	8
40T	30	0.8	47.68	7	21.09	6
40T	50	0.2	time	time	100.39	9
40T	50	0.4	time	time	105.85	13
40T	50	0.6	1,255.49	16	43.77	10
40T	50	0.8	125.30	11	26.25	7

time: 6 h time limit exceeded.

computational resources and runtime are increasingly required. For future work, it is suggested to consider uncertainties in the problem parameters, such as uncertain demands and costs. Another important research direction is to apply heuristic and meta-heuristic methods to solve the problem.

CRedit authorship contribution statement

Fabricio Alves Oliveira: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing – original draft, Writing – review & editing, Visualization. **Elisângela Martins de Sá:** Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization, Supervision. **Sérgio Ricardo de Souza:** Conceptualization, Methodology, Data Curation, Writing – original draft, Writing – review & editing, Supervision.

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Table B.12

Comparison of four strategies for adding cuts in the branch and bound tree: Only for integer solutions (BDCFI), only for fractional solutions at the root node (BDCFR), for all integer solutions and fractional solutions at the root node (BDCF) and for all integer and fractional solutions (BDCFA)

Instance	Revenue	α	BDCFI			BDCFR			BDCF			BDCFA		
			Time (s)	Cuts	Nodes	Time (s)	Cuts	Nodes	Time (s)	Cuts	Nodes	Time (s)	Cuts	Nodes
10L	20	0.2	0.69	431	45	0.50	811	0	0.58	531	0	0.59	531	0
10L	20	0.4	0.26	237	0	0.17	423	0	0.26	279	0	0.25	279	0
10L	20	0.6	0.16	172	0	0.08	300	0	0.17	198	0	0.17	198	0
10L	20	0.8	0.15	161	0	0.11	321	0	0.16	161	0	0.15	161	0
10L	30	0.2	0.69	447	30	0.35	699	0	0.53	498	0	0.52	498	0
10L	30	0.4	0.26	231	0	0.22	444	0	0.26	246	0	0.26	246	0
10L	30	0.6	0.16	203	0	0.09	300	0	0.18	246	0	0.18	246	0
10L	30	0.8	0.23	234	0	0.24	551	0	0.27	240	0	0.26	240	0
10L	50	0.2	0.82	466	88	0.27	569	0	0.57	541	0	0.56	541	0
10L	50	0.4	0.27	228	0	0.17	413	0	0.25	240	0	0.25	240	0
10L	50	0.6	0.27	261	0	0.18	446	0	0.27	265	0	0.24	265	0
10L	50	0.8	0.25	249	0	0.21	576	0	0.29	252	0	0.29	252	0
10T	20	0.2	1.01	507	140	0.53	742	0	0.90	612	1	0.89	612	1
10T	20	0.4	0.35	275	0	0.23	557	0	0.38	383	0	0.35	383	0
10T	20	0.6	0.33	210	0	0.31	670	0	0.42	235	0	0.41	235	0
10T	20	0.8	0.16	180	0	0.20	442	0	0.27	197	0	0.27	197	0
10T	30	0.2	0.84	458	20	0.49	718	0	0.69	526	0	0.72	526	0
10T	30	0.4	0.42	349	13	0.29	632	0	0.30	420	0	0.30	420	0
10T	30	0.6	0.49	331	17	0.36	661	0	0.71	444	1	0.69	444	1
10T	30	0.8	0.21	270	0	0.18	502	0	0.24	310	0	0.23	310	0
10T	50	0.2	0.85	498	4	0.50	795	0	0.87	593	0	0.87	593	0
10T	50	0.4	0.46	325	0	0.28	612	0	0.38	412	0	0.36	412	0
10T	50	0.6	0.44	349	14	0.60	740	0	0.67	461	5	0.66	482	1
10T	50	0.8	0.22	280	0	0.14	399	0	0.24	324	0	0.23	324	0
20L	20	0.2	6.07	1,802	59	5.18	4,037	0	5.30	2,516	0	5.32	2,516	0
20L	20	0.4	3.08	903	0	1.91	2,195	0	2.51	1,022	0	2.50	1,022	0
20L	20	0.6	1.35	707	0	1.23	2,000	0	1.36	707	0	1.36	707	0
20L	20	0.8	1.08	685	0	0.92	1,600	0	1.08	685	0	1.08	685	0
20L	30	0.2	27.95	2,467	1130	13.02	5,714	3	9.25	3,389	0	9.36	3,389	0
20L	30	0.4	6.89	1,835	40	4.39	3,413	0	3.95	2,091	0	3.94	2,091	0
20L	30	0.6	3.02	1,254	0	2.22	2,486	0	2.76	1,294	0	2.75	1,294	0
20L	30	0.8	2.06	1,065	0	1.77	2,313	0	2.07	1,065	0	2.08	1,065	0
20L	50	0.2	15.10	2,210	830	16.91	6,657	0	7.43	3,097	0	7.41	3,097	0
20L	50	0.4	10.28	1,989	120	3.55	2,919	0	5.87	2,375	0	5.85	2,375	0
20L	50	0.6	4.41	1,681	0	2.80	3,087	0	3.73	1,717	0	3.75	1,717	0
20L	50	0.8	3.25	1,475	0	2.66	2,910	0	2.89	1,568	0	2.88	1,568	0
20T	20	0.2	4.39	1,369	0	3.05	3,090	0	4.20	1,651	0	4.24	1,651	0
20T	20	0.4	1.92	911	0	1.93	2,800	0	1.93	911	0	1.92	911	0
20T	20	0.6	1.57	781	0	1.20	1,614	0	1.58	781	0	1.58	781	0
20T	20	0.8	1.49	745	0	1.34	1,633	0	1.68	750	0	1.71	750	0
20T	30	0.2	11.88	2,392	0	9.64	4,913	5	8.30	3,363	0	8.33	3,363	0
20T	30	0.4	4.12	1,892	0	3.43	3,267	0	3.41	1,746	0	3.42	1,746	0
20T	30	0.6	3.79	1,352	0	2.87	3,263	0	3.77	1,352	0	3.77	1,352	0
20T	30	0.8	3.52	1,265	7	2.04	2,227	0	3.23	1,364	0	3.25	1,364	0
20T	50	0.2	23.30	2,932	340	5.90	4,096	0	8.35	3,733	0	8.39	3,733	0
20T	50	0.4	7.23	2,251	9	4.51	3,606	0	3.85	2,331	0	3.84	2,331	0
20T	50	0.6	3.56	1,681	0	6.91	3,751	17	3.38	1,784	0	3.36	1,784	0
20T	50	0.8	2.73	1,509	0	1.57	1,977	0	2.86	1,733	0	2.89	1,733	0

(continued on next page)

Appendix A. Complement for the proof of Proposition 1

Let $s_2 = (h, z, y, f)$ be the solution defined in Proposition 1, from a solution $s_1 \in P_1$. The proof that s_2 belongs to P_2 and has the same objective function value as s_1 is given below.

Proposition 2. *Let s_1 and s_2 be the solutions given in Proposition 1. Then the solution s_2 belongs to P_2 and has the same objective function value as s_1 .*

Proof. Let R denotes the set of pairs (k, m) such that k and m are, respectively, the first and last hub of a route used to serve the pair (i, j) , as defined in Proposition 1. To verify that $s_2 \in P_2$, it is just need to show that it satisfies the constraints (16)–(22). By construction, the constraints (16), (21) and (22) are satisfied. The set of constraints (17) is also satisfied, since:

$$\sum_{k \in N} \sum_{m \in N} y_{ijkm} = \sum_{(k,m) \in R} y_{ijkm} = \sum_{k: (k,m) \in R} a_{ijk} = \sum_{k \in N} a_{ijk} \leq 1, \forall i, j \in N.$$

Table B.12 (continued).

Instance	Revenue	α	BDCFI			BDCFR			BDCF			BDCFA		
			Time (s)	Cuts	Nodes	Time (s)	Cuts	Nodes	Time (s)	Cuts	Nodes	Time (s)	Cuts	Nodes
25L	20	0.2	14.70	3,029	50	7.15	4,145	0	15.40	3,864	0	15.50	3,864	0
25L	20	0.4	7.52	1,844	0	6.85	4,963	0	6.23	1,866	0	6.27	1,866	0
25L	20	0.6	5.49	1,289	0	3.12	2,826	0	5.51	1,289	0	5.46	1,289	0
25L	20	0.8	1.91	1,073	0	1.47	1,875	0	1.88	1,073	0	1.92	1,073	0
25L	30	0.2	60.53	3,988	320	14.88	6,650	0	14.77	4,670	0	14.78	4,670	0
25L	30	0.4	11.27	2,631	29	6.63	4,754	0	10.44	3,043	0	10.31	3,043	0
25L	30	0.6	6.64	2,273	0	6.26	5,017	0	7.81	2,286	0	7.80	2,286	0
25L	30	0.8	7.68	1,880	13	4.46	4,119	0	5.60	2,085	0	5.62	2,085	0
25L	50	0.2	34.68	4,037	720	19.17	6,884	21	15.01	5,108	0	15.13	5,108	0
25L	50	0.4	24.02	3,415	140	7.99	5,129	0	11.61	3,827	0	11.67	3,827	0
25L	50	0.6	9.08	2,701	0	5.15	4,008	0	10.06	3,081	0	10.07	3,081	0
25L	50	0.8	6.50	2,344	0	4.03	3,460	0	6.84	2,710	0	6.85	2,710	0
25T	20	0.2	5.52	1,923	0	3.92	3,421	0	5.64	1,923	0	5.55	1,923	0
25T	20	0.4	4.28	1,396	0	2.81	2,640	0	4.28	1,396	0	4.31	1,396	0
25T	20	0.6	2.48	1,225	0	2.09	2,500	0	2.47	1,225	0	2.49	1,225	0
25T	20	0.8	2.35	1,181	0	1.96	2,500	0	2.36	1,181	0	2.37	1,181	0
25T	30	0.2	16.31	3,568	0	14.68	6,692	0	15.28	4,695	0	15.32	4,695	0
25T	30	0.4	7.96	2,688	0	5.64	4,610	0	8.21	2,909	0	8.25	2,909	0
25T	30	0.6	8.41	2,415	0	6.66	4,395	0	8.97	2,516	0	8.92	2,516	0
25T	30	0.8	6.63	1,939	0	4.83	3,547	0	7.64	2,132	0	7.64	2,132	0
25T	50	0.2	23.14	4,276	30	17.30	7,868	0	18.16	5,436	0	18.09	5,436	0
25T	50	0.4	14.03	3,518	0	7.08	5,174	0	8.90	3,496	0	8.97	3,496	0
25T	50	0.6	6.16	2,707	0	4.49	3,962	0	8.36	3,135	0	8.33	3,135	0
25T	50	0.8	4.96	2,302	0	3.43	3,132	0	5.34	2,589	0	5.35	2,589	0
40L	20	0.2	154.86	8,074	65	58.35	14,171	0	106.43	10,034	0	105.57	10,034	0
40L	20	0.4	56.13	5,145	0	34.67	11,186	0	48.46	5,391	0	48.17	5,391	0
40L	20	0.6	22.48	3,964	0	20.11	9,600	0	22.70	3,964	0	22.42	3,964	0
40L	20	0.8	14.87	3,432	0	12.22	6,400	0	14.91	3,432	0	14.85	3,432	0
40L	30	0.2	400.66	13,168	370	106.70	17,950	0	225.34	17,440	0	223.30	17,440	0
40L	30	0.4	93.70	7,811	20	53.66	13,743	0	76.65	9,225	0	76.92	9,225	0
40L	30	0.6	57.14	6,988	0	40.06	13,325	0	52.77	7,110	0	52.46	7,110	0
40L	30	0.8	33.25	5,176	0	27.32	9,528	0	30.05	5,604	0	29.88	5,604	0
40L	50	0.2	894.34	18,079	880	74.92	14,989	0	179.74	16,977	0	178.54	16,977	0
40L	50	0.4	274.10	12,544	329	97.30	18,443	0	99.65	13,545	0	98.84	13,545	0
40L	50	0.6	81.68	8,962	18	49.27	14,060	0	66.16	9,493	0	65.97	9,493	0
40L	50	0.8	57.90	7,812	0	28.33	10,303	0	44.22	7,456	0	43.59	7,456	0
40T	20	0.2	42.61	5,814	0	35.10	10,979	0	43.04	6,076	0	42.72	6,076	0
40T	20	0.4	25.07	4,433	0	22.51	9,600	0	25.23	4,433	0	24.97	4,433	0
40T	20	0.6	19.20	3,868	0	16.54	8,000	0	19.29	3,868	0	19.27	3,868	0
40T	20	0.8	17.67	3,710	0	15.17	8,000	0	17.74	3,710	0	17.66	3,710	0
40T	30	0.2	81.53	9,561	0	55.02	13,927	0	75.47	9,677	0	74.99	9,677	0
40T	30	0.4	40.77	6,891	0	39.19	12,923	0	41.00	6,891	0	40.81	6,891	0
40T	30	0.6	40.08	6,537	0	30.78	10,946	0	40.29	6,537	0	40.23	6,537	0
40T	30	0.8	29.48	5,871	0	20.66	8,472	0	29.63	5,871	0	29.44	5,871	0
40T	50	0.2	145.96	12,099	40	77.54	16,856	0	96.44	13,103	0	95.37	13,103	0
40T	50	0.4	69.78	9,853	4	55.77	15,683	0	69.71	10,448	0	69.32	10,448	0
40T	50	0.6	46.16	8,165	2	49.38	16,954	0	41.79	8,792	0	41.49	8,792	0
40T	50	0.8	41.34	7,216	0	32.36	12,434	0	40.43	7,716	0	40.21	7,716	0

For constraints (18), there are four cases to consider. Given $i, j, k \in N$, we have that if the demand of the pair (i, j) is met, then it will occur along a path where the hub k : (i) it is the only hub on the path; (ii) is the first hub in the path; (iii) is the last hub in the path; (iv) does not belong to the path. If case (i) or (ii) occurs, we have to:

$$\sum_{m \in N} y_{ijkm} + \sum_{\substack{m \in N \\ m \neq k}} y_{ijmk} = \sum_{m \in N} y_{ijkm} + 0 = \sum_{m: (k,m) \in R} y_{ijkm} = a_{ijk} \leq h_k.$$

If case (iii) occurs, then:

$$\sum_{m \in N} y_{ijkm} + \sum_{\substack{m \in N \\ m \neq k}} y_{ijmk} = 0 + \sum_{\substack{m \in N \\ m \neq k}} y_{ijmk} = \sum_{\substack{m: (m,k) \in R \\ m \neq k}} y_{ijmk} = b_{ijk} \leq h_k.$$

Finally, if case (iv) occurs, then

$$\sum_{m \in N} y_{ijkm} + \sum_{\substack{m \in N \\ m \neq k}} y_{ijmk} = 0 \leq h_k.$$

Hence, the set of constraints (18) is satisfied by s_2 .

To verify that s_2 satisfies the constraints (19), see that, by hypothesis,

$$a_{ijk} + \sum_{\substack{m \in N \\ m \neq k}} x_{ijmk} = b_{ijk} + \sum_{\substack{m \in N \\ m \neq k}} x_{ijkm}, \quad \forall i, j, k \in N.$$

Therefore, for all $i, k \in N$, then

$$\sum_{j \in N} w_{ij} \left(a_{ijk} + \sum_{\substack{m \in N \\ m \neq k}} x_{ijmk} \right) = \sum_{j \in N} w_{ij} \left(b_{ijk} + \sum_{\substack{m \in N \\ m \neq k}} x_{ijkm} \right).$$

that is,

$$\sum_{j \in N} w_{ij} a_{ijk} + \sum_{j \in N} \sum_{\substack{m \in N \\ m \neq k}} w_{ij} x_{ijmk} = \sum_{j \in N} w_{ij} b_{ijk} + \sum_{j \in N} \sum_{\substack{m \in N \\ m \neq k}} w_{ij} x_{ijkm}.$$

By (24), (25) and (26), then

$$\sum_{j \in N} \sum_{m \in N} w_{ij} y_{ijkm} + \sum_{\substack{m \in N \\ m \neq k}} f_{imk} = \sum_{j \in N} \sum_{m \in N} w_{ij} y_{ijmk} + \sum_{\substack{m \in N \\ m \neq k}} f_{ikm}, \quad \forall i, k \in N.$$

For the set of constraints (20), can be seen that

$$f_{ikm} = \sum_{j \in N} w_{ij} x_{ijkm} \leq \sum_{j \in N} w_{ij} z_{km}, \quad \forall i, k, m \in N, k \neq m,$$

since, by hypothesis, $x_{ijkm} \leq z_{km}$, for all $i, k, m \in N, k \neq m$.

To verify that the value of the objective function produced by s_2 is equal to the value obtained by s_1 , just note that the following equalities are valid:

$$\begin{aligned} \sum_{i \in N} \sum_{j \in N} \sum_{k \in N} r_{ij} w_{ij} a_{ijk} &= \sum_{i \in N} \sum_{j \in N} \sum_{k \in N} r_{ij} w_{ij} \left(\sum_{m: (k,m) \in R} y_{ijkm} \right) \\ &= \sum_{i \in N} \sum_{j \in N} \sum_{k \in N} \sum_{m \in N} r_{ij} w_{ij} y_{ijkm}, \end{aligned}$$

$$\begin{aligned} \sum_{i \in N} \sum_{j \in N} \sum_{k \in N} c_{ik} w_{ij} a_{ijk} &= \sum_{i \in N} \sum_{j \in N} \sum_{k \in N} c_{ik} w_{ij} \left(\sum_{m: (k,m) \in R} y_{ijkm} \right) \\ &= \sum_{i \in N} \sum_{j \in N} \sum_{k \in N} \sum_{m \in N} c_{ik} w_{ij} y_{ijkm}, \end{aligned}$$

$$\begin{aligned} \sum_{i \in N} \sum_{j \in N} \sum_{m \in N} c_{mj} w_{ij} b_{ijm} &= \sum_{i \in N} \sum_{j \in N} \sum_{m \in N} c_{mj} w_{ij} \left(\sum_{k: (k,m) \in R} y_{ijkm} \right) \\ &= \sum_{i \in N} \sum_{j \in N} \sum_{k \in N} \sum_{m \in N} c_{mj} w_{ij} y_{ijkm} \end{aligned}$$

and

$$\begin{aligned} \sum_{i \in N} \sum_{j \in N} \sum_{k \in N} \sum_{m \in N, m \neq k} \alpha c_{km} w_{ij} x_{ijkm} &= \sum_{i \in N} \sum_{k \in N} \sum_{m \in N, m \neq k} \alpha c_{km} \left(\sum_{j \in N} w_{ij} x_{ijkm} \right) \\ &= \sum_{i \in N} \sum_{k \in N} \sum_{m \in N, m \neq k} \alpha c_{km} f_{ikm}. \end{aligned}$$

Therefore, the solution s_2 belongs to P_2 and has the same objective function value as s_1 . The proof is finished $\square$

Appendix B. Detailed tables of results

See Tables B.10–B.12

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